
INTEGRATED SMART CAMPUS VEHICLE MANAGEMENT SYSTEM: RFID ACCESS, VIOLATION TRACKING, AND AI-POWERED PARKING OPTIMIZATION FOR CSPC

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CHAPTER 1

INTRODUCTION

As institutions expand, controlling the movement of people and vehicles becomes more difficult. Conventional practices may have a hard time catching up to the speed of today's campuses, causing vulnerabilities in terms of safety and hassles for all the people working and studying within. This chapter will present a smarter approach that will revolutionize entry and parking in CSPC.

1.1 Project Context

Across the world, the rise in numbers of urban population and motorized vehicles has posed a problem in terms of traffic control and parking facilities. It is stated by the International Energy Agency (IEA) that there were more than 1.45 billion vehicles globally in the year 2023, and the figures keep increasing every year. The rise in numbers has led to problems such as traffic jams, ineffective parking facilities, and security concerns related to vehicle surveillance [1]. Research indicates that on average, drivers use about 20–30 percent of their driving time looking for parking spaces, contributing to traffic pollution and carbon emission. Consequently, many countries have turned towards implementing Smart City technology that involves digitalization using IoT, AI, and automation.

Other important measures that have been implemented under the concept of Smart Cities include the use of smart parking services and access control mechanisms for the vehicles. It has been observed that the implementation of such smart parking measures can help save 40 percent of the search time needed for identifying vacant spots and can prove beneficial for improving the efficiency of traffic management in controlled environments such as university campuses, malls, and central business districts. Such smart parking and access control systems usually employ technologies such as RFID and sensors for vehicle detection. Due to their efficient and secure

nature, several universities around the globe are adopting the Smart Campus concept by incorporating digital technologies in their campus infrastructure.

The increase in the number of registered vehicles in the Philippines is also contributing to more complex traffic and parking management. According to statistics provided by the Land Transportation Office (LTO), the total number of registered motor vehicles in the country was already more than 13 million by 2023 [9]. Given the fast-paced changes happening because of urbanization and technological advancement, various government agencies, including the Department of Science and Technology (DOST) and the Commission on Higher Education (CHED), are encouraging educational institutions to adopt smart technologies in their systems and apply digital management through monitoring and access control systems. Despite all these steps taken by the government, many higher learning institutions, particularly those found in the provinces, still use traditional vehicle tracking devices like stickers and logbooks.

The entry and parking of vehicles have always been controlled by a manual system involving ID stickers and log books recorded by university security guards in Camarines Sur Polytechnic Colleges. Despite being used for several years now, the manual control of entry and parking poses a number of problems. For instance, during rush hours, when there is heavy traffic, it involves the need for security officers to check manually the ID stickers of vehicles and keep track of parking inside the university campus. Moreover, without an automatic parking system that keeps track of empty parking lots, drivers are forced to look for open parking spaces manually.

From the first investigation done by the researchers at the GSU and campus security staff, it is found out that there are increasing numbers of cars entering the CSPPC premises on a daily basis as the population of students, teachers, and other employees are increasing. Nevertheless, the college is yet to have any sort of advanced system to manage and monitor the vehicles as the manual way of doing so is quite tedious for both the college management and the campus security officers. Therefore, an advanced vehicle management system should be introduced for the purpose of better

vehicle security and management. This study therefore proposes the design and development of a smart vehicle management system for CSPC using RFID and artificial intelligence technologies.

1.2 Purpose and Description of the Project

In summary, the objective of this study project is to come up with an Integrated Smart Campus Vehicle Management System (ISCVMS) for Camarines Sur Polytechnic Colleges (CSPC). The ISCVMS is designed to automate and optimize the entry and exit of vehicles from the school grounds, as well as secure and organize their parking within the school grounds. The ISCVMS seeks to improve upon the current manually operated method of managing the entry and exit of vehicles in which stickers and logbooks are used to monitor their movement.

The working of this system relies on the integration of several systems into one integrated platform. Firstly, the RFID technology is implemented to recognize the identity of the authorized vehicles when entering the campus through the entrance gates. Secondly, an intelligent module is used for parking detection using cameras that can help in determining the number of parking spaces available. Thirdly, this system will have a central computer-based database which will record all activities of the cars including violation of parking rules and send notification automatically via SMS or emails. The system uses artificial intelligence primarily for detecting vehicles within the parking area. The determination of whether a vehicle is properly parked or violating parking rules is handled through rule-based analysis using predefined parking slot boundaries. The significance of this project lies in the fact that it can make the work of the campus administration and security officials easier by providing them with all data in just one screen. It becomes valuable in terms of improving the efficiency of the operation 4 process on campus and ensuring better security. The latter can be achieved by minimizing the traffic load on campus and removing the need for manual registration of cars' data.

This project is also innovative in that it brings together the three technologies of IoT-enabled RFID gate entry control, artificial intelligence for detecting parked vehicles, and the central

monitoring system of parking violations into one platform. Whereas most other systems concentrate solely on either parking or access control, this project utilizes several smart systems to develop an efficient means of managing vehicles on campus. With this system in place, CSPC will move closer to being a smart campus by creating a safer and technologically advanced environment for its stakeholders.

1.3 Objectives of the Project

The overall goal of this research is to create an Integrated Smart Campus Vehicle Management System for Camarines Sur Polytechnic Colleges (CSPC) through the use of Agile Software Development Methodology. Using this methodological approach, the processes of iteration are facilitated so that there can be efficient and effective creation of automated vehicle access, artificial intelligence-driven parking security, and digitization of violation tracking systems.

Specifically, this study aims to:

1. Perform a thorough review of the existing manual procedures at CSPC in order to define the necessary elements of the system along with the system's technical requirements.
2. Develop a web-based system that allows the following:
 - 2.1. User management module;
 - 2.2. Vehicle registration module;
 - 2.3. RFID gate access module;
 - 2.4. AI-Powered parking monitoring module;
 - 2.5. Violation tracking module;
3. Evaluate the system using BBT (blockbox testing) in terms of;
 - 3.1. Functionality;
 - 3.2. Usability;
 - 3.3. Reliability;

- 3.4. Security;
- 3.5. Performance;
- 4. Implement and deploy the system within Camarines Sur Polytechnic Colleges;

1.4 Significance of the Study

The researchers think that implementation of an Integrated Smart Campus Vehicle Management System with RFID Access, Violation Tracking, and AI-Powered Parking Optimization at CSPC will enhance efficiency and management of entry as well as parking of vehicles on campus. The benefits from the findings of this research would include:

Camarines Sur Polytechnic Colleges (CSPC). A modern vehicle control system will be useful for the institution, making it easier to organize and secure the campus. The use of an RFID access system and automated monitoring will go a long way in minimizing the unauthorized use of vehicles and improving traffic flow on campus.

Campus Security Personnel. This system is expected to help the security staff in terms of minimizing the logging process as well as checking for stickers. Vehicle recognition and violation registration processes become automated and will enable the guards to concentrate on keeping things orderly.

Students. With respect to students utilizing motorized vehicles in the university campus, the new system will increase their convenience through quick access to the gates and availability of parking space information.

Personnel. The proposed system will aid campus administrators through providing an effective process of vehicle management and access control. The quick response in accessing vehicles as well as parking availability will minimize the delays experienced in the campus.

Researchers. The purpose of this research is to provide an opportunity for the researchers to utilize the theoretical and practical aspects of their learning in the field of Information

Technology by creating a system combining RFID, Internet of Things, and Artificial Intelligence-based parking detection systems.

Future Researchers and Developers. This paper can be used by other researchers in future research that is related to the construction of smart campuses. This research offers information on how emerging technologies can be incorporated to provide solutions to parking lots and vehicle management.

1.5 Scope and Limitations of the Project

The study shall aim to design and develop an Integrated Smart Campus Vehicle Management System in Camarines Sur Polytechnic Colleges. In order to create the necessary system design, there will be a need to assess the current manual process used by the institution in monitoring vehicles in terms of problems that may hinder their daily functions, and these shall serve as bases for the product backlog. The proposed web application shall include several modules, including User Management Module, which is meant for managing user accounts with specific roles; the Vehicle Registration Module, which is responsible for keeping track of authorized vehicles; the RFID Gate Access Module, which involves automated entrance and exit of the vehicle through the use of RFID; the AI-Powered Parking Monitoring Module, wherein computer vision technologies are utilized in tracking down open and filled parking spots; and the Violation Tracking Module, where violations regarding parking and traffic are tracked down inside the campus. In checking whether the system passes the standard requirements, the researchers will implement Black Box Testing (BBT) in terms of functionality, usability, reliability, security and performance.

Yet, there are a few shortcomings of the system. First, the RFID element only authenticates the RFID tag installed on the vehicle but fails to authenticate the identity of the driver. Hence, the tag can be used by another individual in case of transferring it to other individuals. Second, the artificial intelligence-based car detection system might be impacted by external factors like

insufficient illumination, weather interference, or blockage of cameras. Moreover, testing and deploying the proposed system are restricted to certain parking lots of CSPC College only.

1.6 Project Dictionary

This section defines key technical and operational terms used throughout the study to ensure clarity.

Artificial Intelligence (AI). It is referred to as a branch of computer science which entails modeling of intelligence by computers in regard to their ability to imitate intelligent functions in human beings, like vision and reasoning [4]. In this research, the purpose of the paper is explained as the implementation of the YOLOv9 model for detecting free and occupied parking spots through analysis of live videos from the CSPC parking lots.

Blind Circling. It is an inefficient way of driving in which drivers keep going around a confined area in search of a specific resource such as parking space without prior knowledge of its availability [6]. This study refers to the unnecessary fuel wastage and time taken by CSPC drivers while looking for available parking slots.

Black Box Testing (BBT). It is a software testing method that evaluates the functionality of a system by examining inputs and outputs without considering the internal program structure. In this study, BBT is used to evaluate the proposed system in terms of functionality, usability, reliability, security and performance.

Computer Vision. It is a subset of artificial intelligence that enables computers and systems to “recognize and understand visual information from images and videos” [6]. In this study refers to the software's capacity to recognize and understand the CSPC parking lot with the help of camera sensors and determine if there is a vehicle present within the designated area.

Dwell Time. The time taken by an individual or system to stay in a specific state or location within a particular process. This study refers to the time taken by an automobile to stay at the CSPC gate

while undergoing security checks. The aim is to minimize this time with the help of an RFID-based system.

Functionality. It refers to the degree to which a software system provides functions that meet specified requirements and user needs. In this study, functionality measures whether the proposed vehicle management system performs its intended operations correctly and efficiently.

Internet of Things (IoT). Defined as a group of interconnected computing devices, mechanical devices, or objects that are assigned a unique identifier, enabling it to transfer data over a network without the need for human intervention [3]. For this study, IoT refers to the group of ESP32 microcontrollers that connect the physical gate hardware and parking cameras to the system's central web-based database.

Performance. It refers to the responsiveness, speed, and stability of a system under a specific workload. In this study, performance measures the ability of the system to process vehicle monitoring transactions efficiently during different levels of campus traffic.

Radio Frequency Identification (RFID). It is described as an electronic system that uses radio waves to transmit information from an object tagged with an RFID tag to an electronic reader device used for identification purposes [8]. In this study, it is described as the device used in the automatic operation of the entrance/exit barrier of CSPC students/staff.

Reliability. It refers to the ability of a system to consistently perform its intended functions without failure over a period of time. In this study, reliability evaluates the consistency and stability of the proposed vehicle management system during continuous operation.

Security. It is the security that protects a system and the information it contains from any form of illegal access, exploitation, or tampering. In this research, the security test assesses how the system secures user accounts, vehicle information, and access controls.

Smart Campus. It is stated that it is a system which involves the use of technology by the school to enhance the quality of education, increase the efficiency of learning places and ensure safety [7].

In this research, the term 'Smart Campus' implies integration of the VMS into the CSPC environment to convert security logging into digital form.

Usability. It means the extent to which the system is easy to learn and use effectively by its intended users. The measure of usability in this case is the level of convenience and satisfaction that the system generates among the administrators, security officers, and vehicle owners.

Vehicle Management System (VMS). It is an integrated software solution that seeks to effectively and comprehensively manage and control various vehicular activities while providing information about vehicular locations and usages to enhance vehicular efficiencies [2]. In this study refers to the integrated web-based software developed to effectively manage the CSPC parking lot.

Violation Tracking. The systematic process of monitoring, identifying, and recording infractions of established rules or regulations in a controlled environment for enforcement purposes [5]. In this study, Violation Tracking refers to the digital module that records instances of illegal parking or speeding at CSPC, implementing the 'Three-Strike Logic' for administrative warnings.

YOLOv9 (You Only Look Once). YOLOv9 is an advanced and powerful software tool that enables the detection and recognition of objects within images and videos with the help of deep learning algorithms [4]. In this study, the YOLOv9 software in this study is used to differentiate between cars, motorcycles, and empty pavement within the CSPC parking area.

Notes

- [1] Soufiene Djahel, Ronan Doolan, Gabriel Miro Muntean, and John Murphy. 2015. A Communications-Oriented Perspective on Traffic Management Systems for Smart Cities: Challenges and Innovative Approaches. *IEEE Communications Surveys and Tutorials* 17, 1 (January 2015), 125–151. <https://doi.org/10.1109/COMST.2014.2339817>
- [2] Chester C Dominic Castro, Mark P Glyn Evans, Janell D Fernando, Christian Paul Reyes, Ian B Carlos Lajom, Claire N Lizarondo Adviser, and Ana B Lizel Del Poso. RFID GATE MONITORING SYSTEM FOR BASIC EDUCATION OF BALIUAG UNIVERSITY WITH SMS NOTIFICATION.
- [3] Hemanth Vijay Reddy Karri. 2025. Next-Generation Smart Parking Systems: A Hybrid Framework for Secure Access and Real-Time Analytics. Retrieved May 14, 2026 from <https://scholarworks.calstate.edu/downloads/8p58pp78r>
- [4] Jing Leng and Yu Guo. 2024. AI-Powered Smart Parking Management: Optimizing Allocation and Safety. *ACM International Conference Proceeding Series* 1, (December 2024), 397–403. <https://doi.org/10.1145/3697355.3697421>;JOURNAL:JOURNAL:ACMOTHERCONFERENCE;CTYPE:STRING:BOOK
- [5] Hao Xuan Nguyen and Minh N.T. Nguyen. 2024. Automated Traffic Violation Monitoring and Reporting System. *Lecture Notes on Data Engineering and Communications Technologies* 229, (2024), 216–226. https://doi.org/10.1007/978-3-031-75593-4_20
- [6] Vijay Paidi, Hasan Fleyeh, Johan Håkansson, and Roger G. Nyberg. 2018. Smart parking sensors, technologies and applications for open parking lots: A review. *IET Intelligent Transport Systems* 12, 8 (October 2018), 735–741. <https://doi.org/10.1049/IET-ITS.2017.0406>;WEBSITE:WEBSITE:IETRESEARCH;WGROU:STRING:PUBLICATION

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- [7] Toni Perković, Pejo Šolić, Hamid Zargariasl, Duje Čoko, and Joel J.P.C. Rodrigues. 2020. Smart Parking Sensors: State of the Art and Performance Evaluation. *J. Clean. Prod.* 262, (July 2020), 121181. <https://doi.org/10.1016/J.JCLEPRO.2020.121181>
- [8] Aep Setiawan, Aep Setiawan, Bayu Widodo, Inna Novianty, Irmansyah Irmansyah, Ridwan Siskandar, Gema Parasti Mindara, Faldiena Marcellita, Dodik Ariyanto, and Lathifunnisa Fathonah. 2024. Implementation of Smart Parking System with RFID Technology in IPB Vocational School Campus. *International Journal of Progressive Sciences and Technologies* 46, 2 (September 2024), 559–564. <https://doi.org/10.52155/ijpsat.v46.2.6621>
- [9] Philippine Statistics Authority. Retrieved May 14, 2026 from <https://psa.gov.ph/statistics/environment/peenra/node/16840811>

CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter provides a review of the literature and systems relating to the development of the Integrated Smart Campus Vehicle Management System with RFID Access, Violation Tracking, and Artificial Intelligence-Powered Parking Optimization for CSPC. This chapter will delve into the prior studies, theories, and technologies related to smart parking, RFID-based access control for vehicles, artificial intelligence, and IoT monitoring systems. The objectives of this chapter include providing a theoretical background for the study and identifying current technologies and methodologies that can aid in designing the suggested system.

2.1 Related Literature and Systems

The following part of the paper discusses the related literature and existing systems that apply to the proposed Integrated Smart Campus Vehicle Management System. The most important technologies that are involved include RFID technology, artificial intelligence, computer vision, smart parking systems, and IoT-based monitoring systems. The use of these technologies will be done for improving the accessibility of vehicles and their effective parking, while ensuring that there is high security in the entire system.

2.1.1 Challenges in Managing Vehicle Traffic in an Institution

Problems in managing the vehicle movement inside the institution can be defined as the challenges faced by the organization while managing the movement of vehicles into and out of the organization and their parking activities. The rising number of vehicles can lead to vehicle jams, shortage of parking space, delays in passing through gateways, and even security threats. Using traditional ways like log books and stickers for checking in vehicles proves to be tedious, and thus becomes a challenge for the institution to manage the vehicle movement and parking activities

efficiently. This problem is related to the first aim of the project which is to evaluate the manual vehicle movement system of CSPC.

The issue of traffic congestion on educational campuses has become a major problem because of the ever-increasing number of motorized vehicles moving on and off these campuses each day. In a case study performed in a university located in Saudi Arabia, the implications of traffic congestion on congested campuses were investigated. Various methods of collecting traffic information were used to come up with potential ways of solving the problem of traffic congestion on campuses [16].

Human behavior and operational constraints can also influence difficulties related to traffic control. The experiences of traffic officers and the difficulties faced during traffic management operations were examined through a mixed method approach in Bacolod City. Various difficulties identified include arrogance of drivers, lack of discipline from motorists, lack of knowledge on traffic rules, poor weather conditions, rise in number of vehicles, and construction activities on the road [22].

With time, technological innovation is becoming more advanced and can be seen being applied in traffic management problems within organizations and urban cities. An innovative solution that was suggested involved developing a sustainable traffic management system through the incorporation of Internet of Things (IoT) and ITS technology. In the development of the system, IoT and ITS devices, artificial intelligence sensors, and machine learning algorithms were employed in collecting and analyzing real-time traffic information. The results obtained indicate efficiency in traffic prediction, reduction of traffic congestion, and efficient transportation management [26].

Lack of sufficient parking areas and the poor parking management system is another reason for institutional traffic challenges. The use of an Internet of Things Raspberry Pi-enabled Parking Management System was adopted to help drivers easily find empty parking spots through real-time monitoring and GPS location system. The system made use of Raspberry Pi technology, ultrasonic

sensors, Pi cameras, GPS location, Wi-Fi capability, and a mobile app linked to the Blynk Internet of Things server. Findings indicated that the system was able to monitor parking status, update parking data in real time, and increase parking efficiency [19].

In another study, an on-line application that offers parking assistance based on permits was assessed at a university. The research focused on changes in terms of parking behavior, parking availability, enforcement equity, and user satisfaction, pre-and-post use of the application. It was found out that using the application led to improved efficiency in parking, such as having less trouble finding parking spots and fully occupied parking zones [8].

Overall, based on the reviewed articles, an increase in vehicle volume, traffic congestion, shortage of parking spaces, and ineffective manual monitoring systems highly impact traffic management in institutions. It is evident from the references gathered above that the integration of innovative technologies such as IoT, AI, RFID, and smart parking solutions will help address issues related to vehicle monitoring, traffic movement, parking system, and decision-making. This information is useful for this current research since it provides an insight into how traffic management is currently being handled and the available technological solutions.

2.1.2 IoT-Based Gate Control System

IoT-based Gate Control System is a name given to an advanced access management technology that employs interconnected physical components, as well as internet-based communication to automate gate operations. IoT allows for the connection of devices such as sensors, RFID readers, cameras, and microcontrollers into a unified system that enables effective monitoring and control. As applied in gate control systems, the IoT allows for vehicle identification, opening and closing of gates, and monitoring of both incoming and outgoing vehicles from a particular site. Such technologies are often employed in institutions, residential neighborhoods, and parking sites. The technology fulfills the aim of automating a campus vehicle management system that controls entrance and exit points on campus.

There has been an increase in the need for a safe and flexible access control solution which has led to the development of the smart gates technology. A research carried out in 2025 suggested a Smart Gates system which would incorporate the use of RFID, biometrics and the use of smartphones. This technology would employ the use of sensors, microcontrollers and wireless connectivity technologies to process data in real time. Machine learning technology would also be used in this technology to deter any unauthorized access attempt [9]

There have also been efforts to use automation using IoT technology in monitoring and controlling infrastructures. An experiment was carried out in an IoT-based dam level monitoring system using ultrasonic and environmental sensors to detect water level, temperature, humidity, and moisture for purposes of predictive analysis and automatic control of gates. The technology used WiFi-enabled microcontrollers to send information detected to the cloud server for monitoring and analysis [32].

Problems encountered in manual operation of systems have also underscored the need for automation and centralized control of gates. In one study, a floodgate management system using IoT technology was proposed in order to solve inefficiencies related to the manual operation of floodgates, especially in remote rural settings where coordination had been challenging. The use of the IoT floodgate management system allowed for automation of opening and closing of floodgates depending on water levels, as well as sending information about the gate operations back to the central IoT network [17].

IoT technology has been adopted together with security and safety systems in automatic gate operations. An automatic gate system with IoT features designed in 2024 included the ESP32 microcontroller that had Wi-Fi capability and could enable remote access through mobile phones. RFID technology was used as an alternative to authenticate entry where internet connections could be lost. For safety purposes, ultrasonic sensors were added to enable obstacle detection as the gates operated, while electromagnetic locks enhanced security once the gates were closed [12].

Regarding safety applications in transport, automated gates have been deployed to enhance traffic safety and reduce accidents. A study on automated railway gate control was developed using LoRa communication modules, IR sensors, ultrasonic sensors, and Arduino Uno. Real-time communication was established between the transmitter and the receiver modules for automated train detection and gate control. Results indicated that the implementation of automatic gate control reduced the safety risks involved in manual gate control [29].

On the whole, it is evident that the examined literature sources confirm the significant superiority of IoT-controlled gate systems over conventional manual ones in terms of security, safety, control, and efficiency. Based on the provided references, it can be stated that the implementation of such technologies as RFID, sensors, remote monitoring via cloud platforms, wireless communications, and control systems proves to be effective in ensuring the appropriate management of gate actions. This literature is useful for the current study because it provides solid evidence supporting the development of an Integrated Smart Campus Vehicle Management System for CSPC.

2.1.3 AI-Powered Parking Detection

AI-Powered Parking Detection System can be defined as an intelligent parking management system which makes use of AI, computer vision, and monitoring technologies to detect occupied and unoccupied parking spaces. In the proposed system, CCTV cameras, sensors, and computing machines are used to capture and analyze the image feeds in order to detect parking status of vehicles. YOLO, Faster R-CNN, and DeepSORT are some of the computer vision techniques which can be used in the detection process. Networking can also be used in order to transfer live video feeds. The proposed system can serve the purpose of this research which aims at designing an intelligent parking management system capable of detecting available parking slots within CSPC.

The increasing difficulties in the parking management process have motivated the implementation of parking technologies powered by artificial intelligence. One investigation carried out in the year 2024 analyzed the impact of artificial intelligence on parking management systems that do not involve highly advanced infrastructure. The application of several artificial intelligence processes including parking slot detection and License Plate Recognition was examined to determine its role in the identification of vehicles and tracking their parking status[18].

AI has also been employed in the detection and tracking of vehicles in real-time in parking scenarios. For instance, a research paper presented a mechanism for automatic tracking using AI whereby YOLO-NAS was used to detect objects, while DeepSORT was used to track the vehicles using PyTorch. The proposed method aimed at detecting the movement of vehicles and identifying the parked vehicles within the urban parking zones. The proposed method proved effective in terms of detecting vehicles and improving the tracking process [5].

Modern parking solutions are equally focused on issues of security and resource optimization. In one of the studies, the disadvantages of conventional parking systems such as inefficiency, lack of monitoring capabilities, and data security problems were identified. As a way to solve these problems, the scholars suggested an advanced parking management system utilizing artificial intelligence, blockchain technology, and IoT technologies. The machine learning algorithms were used for optimization of parking resources, IoT was responsible for monitoring while the blockchain technology facilitated secure data exchange. As a result, the system demonstrated higher efficiency in solving parking problems [23].

The incorporation of AI, IoT, and automation technologies has brought more improvements to parking and traffic management operations. One research done in 2025 was about the implementation of a smart parking system that relied on real-time information collected through IoT sensors for parking availability, dynamic space management, automatic payment processing, and traffic management. Efficiency was the major factor used by the researchers when testing whether the system was useful in enhancing the performance of parking operations in an urban

setting. The findings were that the use of AI and IoT brought improvements to parking management[34].

Parking slot identification systems and vehicle authentication systems have also been developed to solve problems with parking congestion in today's cities. For example, a study proposed a smart parking system using RFID technology for vehicle authentication and Faster R-CNN algorithm for identifying empty parking slots. This system also involved monitoring the time in and out of parked vehicles and notifications regarding long parking duration. From the findings, it was observed that the use of AI, RFID, and computer vision technologies helped enhance parking efficiency and reduce parking congestion.

In conclusion, from the reviewed research papers, it is evident that there has been significant improvement in the field of parking monitoring, vehicle tracking, minimizing traffic congestion, and maximizing the use of available parking spaces through artificial intelligence-based detection systems. These references showcase the efficiency of combining computer vision algorithms like YOLO-NAS, DeepSORT, and Faster R-CNN with IoT and RFID technologies in the process of implementing intelligent parking systems. The discussed results are important to the current research since they assist in designing an efficient smart parking system for CSPC.

2.1.4 Security and Access Control in Smart Gate System

Access control and security are two vital features of smart gates that ensure only authenticated people can enter or leave a secured zone. Smart gates make institutions secure through the authentication process of controlling access permissions. Concerning IT hardware infrastructure, smart gate systems are made up of hardware components including RFID readers, CCTV, biometric scanners, and electronic locks. In software infrastructure, there are databases where information about people's login details, car registration details, and access permissions are stored. Regarding network infrastructure, there is technology for communicating real-time updates, monitoring and

controlling the processes of accessing the smart gate systems. This satisfies the study objective of increasing the security of the institution as well as the automatic access control of vehicles at CSPC.

The development of access control technology has been made possible by the combination of several technologies to achieve optimal security in such facilities. According to research carried out in 2024, License Plate Recognition (LPR), RFID, biometric authentication, and mobile access systems can be combined to provide better security and effectiveness. The access control systems relied on a web-based interface through which security staff controlled access, as well as IP cameras and Python programming techniques to capture license plates in real-time. Other machine learning techniques were also utilized. The result indicated a marked decline in cases of access by unauthorized persons [1] Furthermore, Another technology that has gained popularity is biometrics, which is employed in automated access control systems. In one research study, an automated access control and attendance system was designed using facial recognition and fingerprint authentication to increase efficiency. This process involved controlling gates using Arduino microcontrollers and storing records in MySQL databases. The results obtained from this study indicated increased accuracy in verification and improved security compared to traditional paper and ID card-based systems [21].

The increasing demand for standardization of smart gate designs has also been tackled through reviews of systematic studies. In particular, a systematic review conducted using the PRISMA approach considered 52 articles from 2016 to 2023 to evaluate different types of authentications in smart gates. This systematic review found out that RFID tags, biometric authentication, and mobile phone authentication form the main parts of modern-day access control systems. It was discovered that the use of multi or single factor authentication was critical to improving security levels especially in risky settings [4].

Integrated surveillance and authentication systems have been invented in order to provide additional protection to the vehicles. One of the integrated vehicle surveillance and security systems that used RFID technology was combined with the QR code based visitor authentication, RFID

user verification, and CCTV surveillance system for real time verification. The system included the Raspberry Pi controller which had an access to the MongoDB database where the data related to the users were stored. It has been shown that combining several technologies is efficient for the access control process [28].

Facial recognition software has increased the efficiency of the smart gates through automated identification. Researchers presented a proposed system based on face recognition for use at gates where people were identified through various parameters including lighting, facial expressions, angle, and accessories. Deep learning-based face recognition algorithms helped identify multiple people and grant access to them without any manual intervention once they were recognized. The results showed that the use of face recognition software made access control faster and more efficient [25].

In general, the discussed literature reveals that the use of different kinds of authentication systems, including RFID, biometrics, facial recognition, QR codes, and artificial intelligence-based surveillance systems, would be the most efficient strategy for ensuring the security of and controlling access to smart gates. These systems can considerably increase efficiency, eliminate the possibility of unauthorized access, and provide greater accuracy compared to conventional ways of access control and authentication. In other words, the analyzed literature can be applied to the research since it will contribute to the design of an Integrated Smart Campus Vehicle Management System for CSPC.

2.1.5 SMS Based and Email Notification System

The systems for alerting using either SMS messages or emails are among the critical components of any modern monitoring and management system since these systems offer immediate means of communicating between users of the system and administrators. The notification process is necessary to facilitate quick alerts about critical activities like vehicle entry and exit, availability of parking space, security breach, and many others. In regard to the IT

architecture aspect, hardware such as servers, mobile phones, communication tools is necessary for the transmission of messages. Software will involve the use of databases and application systems to trigger alerts automatically, whereas communication services are used for the purpose of transmitting SMS and emails messages. The network aspect entails having good and stable internet connections and effective communication protocols that enhance the speed of message delivery.

There have been automated alert systems designed with an aim of ensuring reliable and accurate message delivery in monitoring situations. In this regard, one study came up with a design of an alert generation system to help detect messaging problems and guarantee reliable message delivery via SMS and email alerts. The design of the system was done using Python programming language and MySQL for database management. The software used for delivering SMS alerts is known as Twilio, whereas SMTP was used for email alert delivery [2].

In addition to this, machine learning and computer vision have also been incorporated into notification systems for enhancing security monitoring purposes. For instance, one particular study came up with an automatic SMS and email notification system based on the use of the YOLOv9 algorithm. When any kind of intrusion was detected, an immediate notification was sent to predefined individuals through SMS and email notifications. The outcome revealed effective intrusion detection with minimal false alarms [10].

Other communication channels have been integrated through multi-channel communication platforms for better interaction and efficiency within the system. A research study introduced an automation messaging platform that incorporates SMS, e-mails, and messaging apps like WhatsApp to form one platform for efficient communications. The platform was built by employing technologies like React.js for the frontend development, Node.js for backend development, and MongoDB for data storage, while Twilio, Nodemailer, and Puppeteer were used for automating messaging and bulk messaging in file formats respectively [20].

The process of sending alerts automatically by networks monitoring systems has also been perfected over time. A research project conducted a development of a network monitoring system

using the software The Dude on the platform of NDLC, which monitored the failures of devices in a network. Automated notifications were sent via email, SMS, and Telegram in case there were any changes in the state of devices. The results revealed that the messages were transmitted 100 percent successfully with quick responses in all the mediums used [11].

Notifications have similarly been incorporated in academic organizations for purposes of facilitating administrative activities and improving efficiency of communication processes. In one research, a web based student clearance system that incorporated automation of administration process through technologies such as HTML, PHP, CSS, and MySQL was developed. One of the significant modifications done in the system included the incorporation of SMS and Email notifications aimed at communicating student clearance status. This minimized the need for seeking information manually and increased users' awareness [14].

In conclusion, it can be observed from the discussed studies that systems of SMS and email alerts greatly help improve communication efficiency, system responsiveness, and real-time monitoring in different situations. The implementation of an automated alerting system along with AI, IoT, and databases helps deliver information faster and make better decisions based on the received information. This is important for our research paper since it will help us develop an effective system for managing vehicles at CSPC in real-time terms.

2.1.6 System Development Methodology for IoT Based System

Development methodology of IoT systems is crucial since it will provide an effective process of incorporating the different IT components such as hardware, software, and networking. Regarding the IT infrastructure involved, the IoT systems need physical components such as sensors, microcontrollers, RFID tags, and communication components. These need to be effectively designed when coming up with the system. In software aspect, methodologies such as agile and iterative can be used to constantly design and develop new features of the system. Networking wise, development methodology will assist in the selection of suitable communication

protocols and architectures to ensure efficient and effective data transfer between devices in real time. The use of development methodology will ensure that the developed systems are scalable and efficient. The proposed research idea is highly related to this since it aims to design a campus vehicle management system using the Internet of Things concept.

The quick growth of IoT applications in various industries has brought new challenges in dealing with the complexity and interdisciplinarity of their development lifecycle. In one research conducted on this problem, the authors focused on the necessity of the systematic approach to IoT systems development. They studied the lifecycle of IoT applications and developed a conceptual model, including several tasks in the development process. This model was then evaluated by means of a survey among IoT specialists in various fields. It was found that most software developers find IoT development too difficult due to its complexity [13].

The traditional models of software development have also undergone improvement to better meet the demands of developing IoT systems. One study introduced an enhancement to the IoT development process by adding stages in conventional SDLC models to address IoT system development. Some of the new stages included in the enhanced development process model are those related to device requirements definition, communication requirements definition, system integration, and configuration. This enhancement was employed in various fields such as home automation, health care, and environment monitoring [3].

IoT development processes have been streamlined with the development of model-based methodologies. There has been a suggestion of a methodology referred to as the IoT Model Driven Framework (IoTMoF). This framework is designed for supporting requirement analysis, modeling, and code generation based on architectural reference models in IoT systems. It helps in deriving behavioral and configuration models based on the requirements of the IoT systems. This methodology was tested using a smart lighting application [7].

The other technique that can be used for designing IoT systems is MDD. This technique concentrates on making more abstractness and reusability in the system design process. An

investigation was conducted regarding the application of MDD using the Discrete Event System Specification (DEVS) technique to create IoT system models through the hierarchy. This approach enables system simulation and deployment through the integration of models including data acquisition, processing, and networking models together with protocols like MQTT [15].

Moreover, a variety of frameworks using the systems engineering perspective have been created to cope with the complexities of IIoT systems. According to a particular investigation, an IIoT Development Framework that implements principles of systems engineering in its structure, such as business objective identification, system decompositions, and morphological analysis of technologies has been introduced. Such an approach uses IIoT System Canvas for achieving a holistic view on system components, functionalities, and technologies involved in a project [24].

As a whole, it can be said that the studied articles emphasize that the development of IoT systems needs systematic methodologies like Agile, SDLC improvements, MDD, and SE. The above-mentioned approaches enable a better coordination between hardware, software, and network elements while providing opportunities for system scaling and maintenance. It is important to note that the results of the analysis can be applied within the current research since they provide an excellent basis for the development of the suggested Integrated Smart Campus Vehicle Management System for CSPC.

2.1.7 System Testing and Validation for IoT based System

Testing and validation of a system are integral parts of system development in relation to IoT systems. To follow the sequence of activities in testing according to the BSIT curriculum progression, testing will begin with IT Infrastructure Verification to test whether there is correct installation, configuration, and functionality of hardware equipment, sensors, and connectivity modules for effective transmission of real-time data. After this step, the next activity will be Software Design Validation through functional and performance tests to determine whether the application modules such as data processing, user interfaces, and automation functionalities are

operating as per specification. The last step is Network Design and Implementation to verify whether there is smooth integration of communication protocols, data, and responsiveness of different IoT devices with the central server.

Functionality testing is a key step in the development of software as it ensures that the system works as per its designed purpose. As per Ayuningtyas et al., functionality testing involves validating the system features for correctness and generation of the desired output by using black box testing technique where system behavior in terms of inputs and outputs is analyzed, but not the internal coding structure. In their research, tools like Apache JMeter and Katalon Studio were used for performing functional and performance tests of the Perbanas website. The functional testing process included business process analysis, scenario design, automated testing execution, and analysis of outputs, which indicated proper functioning of the system[6]. Similarly, Puji et al. [2024] focused on the importance of testing for functionality while performing a quality assessment of the Supply Chain Management application prototype since all functionalities must function properly without errors prior to implementation. They used both the Equivalence Partitioning method and the ISO/IEC 25010 model to measure the functional suitability and software quality in general, coming up with 254 test cases, wherein 83% of these were valid while 17% were invalid [30].

The purpose of the usability test is to measure the ease, efficiency, and user satisfaction related to a particular software or system designed for its users. The usability of a social networking 16 application was tested by the researchers in Nasyiah et al. [2024] research using the System Usability Scale (SUS). It is one of the most common and widely used tests that use a 5-point Likert scale questionnaire to evaluate user satisfaction and usability. For their study, the researchers used two different scoring techniques, Sauro and Lewis technique and Riihiho technique, in the usability test context. All users carried out predetermined tasks on the application system and rated the usability on the basis of whether they “strongly disagree” or “strongly agree” to statements on usability. Apart from the questions used in the SUS test, qualitative data were gathered using top

down usability testing and interviews to create consistency between the SUS indicator and findings of the usability tests and interviews. From the findings, it can be seen that there was consistency between the results of Sauro and Lewis technique and interviews [27].

Reliability testing, on the other hand, was the topic of discussion in the research by Raksawat et al. [2021], aiming at ensuring that the software is reliable and accurate in its operations during different testing scenarios in accordance with the ISO 29119, which is the principal structure for executing the tests. This structure was used to create a framework for performing the testing procedure, particularly for small businesses, wherein the developers would compare the actual results from users and the expected ones according to the framework.. Black box testing was also conducted to test how reliable the outputs were, without looking at the source codes. User satisfaction levels were gauged using a questionnaire designed by experts, with 78.4% of users reporting satisfaction with the system [31].

The main objective of the security test is to check the ability of the system to ensure protection of users' information and deal with malicious and any other unwanted input. According to Tarune et al. [2025] one of the main criteria when assessing mobile based system for learning discrete mathematics is its security, since the system has to be secure and be able to provide reliable and valid information. While the study mainly used black box testing to determine the efficiency of system performance, the technique was also useful for checking the security of the system through observation of how it responded to unwanted input without the need to reveal internal system structure. In order to assess system performance, black box testing was complemented by conducting a usability evaluation based on questionnaires aimed at obtaining feedback about the system from experts and students. The study found out that the mobile system performed satisfactorily and well-received in terms of usability, thus proving to be efficient and operating safely [33].

Performance testing is done to analyze how efficient, fast, stable, and responsive the system performs with varying workloads. Performance testing has been the center of attention in the study

done by Ayuningtyas et al. because the analysis performed focused mainly on the performance of the system. Performance testing was achieved through automated tests using the black-box testing approach. The researchers have employed the use of Apache JMeter to conduct load and performance testing and Katalon Studio for functional test automation. This included the analysis of the business process, automation of the test scenario and interpretation of the outcomes with regard to the critical performance indicators such as transaction per second (TPS) and response time. As a result, it has been found out that at a load of 250 simultaneous users, with 5,000 test samples for 5 minutes, the average throughput was recorded at 15 TPS, and the failure rate was just 0.06%, which proves the high stability of the system. In addition, the response time varied from 0.2 seconds to 68 seconds, which means that despite a high level of efficiency under heavy loads, some cases should be optimized further. Therefore, the outcome indicates that the system performance is at an acceptable level [6].

In general, it is evident from the reviewed literature that testing and validation of the whole system are essential components that guarantee proper functioning, reliability, and safety of technological systems. In particular, by following specific models, such as ISO/IEC 25010 and ISO 29119, scientists may ensure conformity to international standards that would prove the functional appropriateness and effectiveness of the system. By using automation tools, such as Apache JMeter and Katalon Studio, and Black Box testing, the ability of the system to cope with the load and maintain high accuracy may be checked without looking at the inner structure of the software. Moreover, along with technical validation, user-oriented assessment methods should be used to evaluate the degree to which the technology satisfies users. In particular, the SUS test will help in identifying the level of usability of the system.

2.2 Synthesis of the State-of-the-Art

From the above-reviewed literature and systems, it is clear that there has been a fast evolution in the field of intelligent vehicle management systems with the use of IoT technology, artificial intelligence, RFID-based access control system, and notification systems. As per the

research conducted by Fadlianda et al. [2024], Ibrahim et al. [2024] and Zaheer et al. [2025], these technologies can act as a great platform for building intelligent systems within controlled areas such as campuses.

In the various research articles, it is evident that IoT-powered gate control systems underscore the significance of automation and transmission of data at an instant in order to enhance access management. According to Claver et al. [2025], Fadlianda et al. [2024], and Hajjaj et al. [2020], IoT-powered systems have made it possible for vehicles to enter and leave a building without any physical presence since they incorporate the use of sensors, microcontrollers, and cloud platforms in order to minimize errors. Moreover, according to Patel et al. [2025], the use of RFID technology plays a pivotal role in enhancing access control as they assist in identifying vehicles swiftly.

Concurrently, parking detection through the use of AI technology has proven to be effective in overcoming parking challenges and traffic congestion. As per the study carried out by Ariharan et al. [2024] and S et al. [2025], models based on computer vision like YOLO and faster R-CNN have the ability to identify parking availability and detect vehicles. According to Ibrahim et al. [2024] and Leng et al. [2024], such smart techniques help in effective usage of space and make the life of users more convenient as they reduce their search time within the parking spaces. In addition to it, as mentioned by Zaheer et al. [2025], use of such smart techniques along with IoT makes monitoring possible.

In addition to this, there has been tremendous improvement in the area of security and access control due to the introduction of different types of authentication like RFID cards, biometric devices, and license plate recognition systems. Multi-factor authentication ensures the accuracy of the system as well as minimizes any occurrence of unauthorized access, making it an essential feature in contemporary smart gates. As a follow-up of these characteristics, the importance of SMS and email notifications should not be ignored because these aid in ensuring communication between the user and administrator.

Moreover, research carried out on the methodology behind the design and development of the system highlights the significance of having a well-designed IoT system that is scalable and can be tested easily. According to Alfawair [2022], Boutot [2024], and Minani et al. [2024], the stage in which modeling of the designs and testing are done becomes essential for helping complex systems run efficiently.

It should be noted that based on the results of the state-of-the-art, it can be seen that even though there have been successful applications of some of the individual techniques, like access control, parking detection, and notification mechanisms, the need for integration is growing. According to the studies conducted by Leng et al. [2024] and Zaheer et al. [2025], it would be preferable to integrate the notions of IoT, artificial intelligence, radio frequency identification, and communication within a single framework in order to achieve higher efficiency and effectiveness.

2.3 Gap Bridged by the Study

Despite the many innovative advancements and enhancements in these studies above, there is still a lack in the application of smart vehicular management systems, particularly in university campuses. Most of the existing approaches to creating such smart systems target particular functions like access control, vehicle detection, or notification systems instead of an integrated approach to the development. While one of the papers concentrates on RFID gate automation, another research paper emphasizes the usage of artificial intelligence technology for the purpose of car detection and monitoring. However, these approaches are usually independent of each other and lack the central platform where they can be combined. In addition, the majority of the above discussed systems can be applied exclusively for big cities. They cannot be implemented on college or university campuses, where special adjustments are needed. Management of vehicles in the case of CSPC system is either manual or automated, but only in relation to certain devices such as automatic gates. Moreover, none of the existing models includes vehicle violation tracking and reporting.

To overcome these challenges, the authors have suggested the development of an Integrated Smart Campus Vehicle Management System, which incorporates all these technologies in one platform. In other words, this is a smart system based on access control through RFID technology, parking violation detection using AI, and sending notifications through SMS/emails to notify administrators about violations and their details. Unlike the existing solutions that deal with these issues individually, the Integrated Smart Campus Vehicle Management System makes it possible to monitor the vehicles in real time, effectively manage their activity, and communicate between administrators and users. Thus, the proposed system solves the problem of the need for improvement in previous solutions regarding security, communication, and tracking of violations by students and staff members.

Notes

- [1] Aboagye Isaac Adjaye, Dodoo Ebenezer Nii Darko, Nagai James Tei-Mensah, Kumudzro Selorm Kofi, Freitas Sylvester Nana Yaw, Korang John, Ansah Richardson Margaret, and Sowah Nii Longdon. 2024. Gate Automation Systems Using Access Control Mechanisms. *2024 IEEE 9th International Conference on Adaptive Science and Technology (ICAST)* (October 2024), 1–8. <https://doi.org/10.1109/ICAST61769.2024.10856499>
- [2] Kannan Akshaya and Sanmukh Kaur. 2024. Alert Generation Tool for Messaging Systems. *Development of 6G Networks and Technology* (January 2024), 395–422. <https://doi.org/10.1002/9781394230686.ch18>
- [3] Mai Alfawair. 2022. INTERNET-OF-THINGS: A SYSTEM DEVELOPMENT LIFE CYCLE (SDLC). *J. Theor. Appl. Inf. Technol.* 31, 6 (2022). Retrieved March 23, 2026 from www.jatit.org
- [4] Abdullah M. Almuhaideb, Mariam Elhussein, Reem Osman, Fatema AlHolyal, Leena AlGhamdi, Majd Al-Ismail, Maram Alawami, Zainab Kadour, and Rachid Zagrouba. 2023. Design Recommendations for Gate Security Systems and Health Status: A Systematic Review. *IEEE Access* 11, (2023), 131508–131520. <https://doi.org/10.1109/ACCESS.2023.3335115>
- [5] Mohd Ariharan, Sujitha Juliet, and Janih Anitha. 2024. YOLO NAS-Powered Intelligent Vehicle Tracking and Monitoring System for Enhanced Parking Management in Urban Environments. *10th International Conference on Advanced Computing and Communication Systems, ICACCS 2024* (2024), 2415–2420. <https://doi.org/10.1109/ICACCS60874.2024.10716939>
- [6] Putri Kusuma Ayuningtyas, Dwi Atmodjo WP, and Pratiwi Rachmadi. 2023. Performance And Functional Testing With The Black Box Testing Method. *International Journal of Progressive Sciences and Technologies* 39, 2 (July 2023), 212. <https://doi.org/10.52155/IJPSAT.V39.2.5471>

-
- [7] Paul Boutot. 2024. A Rapid Development Framework for IoT Systems. (August 2024). <https://doi.org/10.32920/26866624.v1>
- [8] Sai Sneha Channamallu, Sharareh Kermanshachi, Jay Michael Rosenberger, Apurva Pamidimukkala, and Greg Hladik. 2025. Smart parking intervention: impact on university mobility. *Transportation Research Procedia* 91, (2025), 401–408. <https://doi.org/10.1016/J.TRPRO.2025.10.052>
- [9] Niyigena Claver, Wilson Musoni, and Niyirora Didace. 2025. Develop an Extended IoT Based Smart Gate System for Vehicles: A Case of Rp-Huye College. *Int. J. Innov. Sci. Res. Technol.* (May 2025), 1872–1883. <https://doi.org/10.38124/ijisrt/25apr1217>
- [10] Eeshaan Dhanuka and P Dhavakumar. 2024. Automated Email and SMS Notification System for Unauthorized Entry using Yolov9. *2024 10th International Conference on Communication and Signal Processing (ICCSP)* (April 2024), 271–276. <https://doi.org/10.1109/ICCSP60870.2024.10544306>
- [11] Moch Khaidar Elhaq, Arip Solehudin, and Didi Juardi. 2021. Penerapan The Dude Sebagai Sistem Monitoring dengan Notifikasi Otomatis Melalui Email, Telegram dan SMS. *Syntax Literate ; Jurnal Ilmiah Indonesia* 6, 7 (July 2021), 3380. <https://doi.org/10.36418/syntax-literate.v6i7.3640>
- [12] David Fadlianda, Muhammad Fikry, and Nunsina. 2024. Innovative IoT-Based Automatic Gate System with RFID and Electro-Magnetic Lock for Secure Access. *Proceedings of Malikussaleh International Conference on Multidisciplinary Studies (MICO MS)* 4, (December 2024), 00005–00005. <https://doi.org/10.29103/micoms.v4i.884>
- [13] Mahdi Fahmideh, Aakash Ahmad, Ali Behnaz, John Grundy, and Willy Susilo. 2022. Software Engineering for Internet of Things: The Practitioners’ Perspective. *IEEE Transactions on Software Engineering* 48, 8 (August 2022), 2857–2878. <https://doi.org/10.1109/TSE.2021.3070692>

-
- [14] Johnson Tunde Fakoya, Ajinaja Micheal Olalekan, Olayemi Olasehinde, and Johnson Olanrewaju Victor. 2021. Design and Implementation of a Web-Based Sms-Notification Clearance System: A Case Study of Federal Polytechnic, Ile -Oluji, Ondo State. <https://doi.org/10.35629/5252-0310942950>
- [15] Iman Alavi Fazel and Gabriel Wainer. 2024. A DEVS-Based Methodology for Simulation and Model-Driven Development of IoT. *Lecture Notes of the Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering, LNICST 519 LNICST*, (January 2024), 3–17. https://doi.org/10.1007/978-3-031-57523-5_1
- [16] Said Ali Hassan, Seraj Yousef Abed, and Wael Salah Hassanein. 2021. Analyzing and Soft Solutions to Vehicle Traffic Problems in Crowded Institutions. <https://services.igi-global.com/resolvedoi/resolve.aspx?doi=10.4018/978-1-7998-8040-0.ch009> (2021), 189–213. <https://doi.org/10.4018/978-1-7998-8040-0.CH009>
- [17] Sami Salama Hussen Hajjaj, Mohamed Thariq Hameed Sultan, Muhammad Hafizuddin Moktar, and Seng Hua Lee. 2020. Utilizing the Internet of Things (IoT) to Develop a Remotely Monitored Autonomous Floodgate for Water Management and Control. *Water (Basel)*. 12, 2 (February 2020), 502. <https://doi.org/10.3390/w12020502>
- [18] Islam M. Ibrahim, Mohamed Khaled, Marwan Ashraf, Mario George, and Iyad Abuhadrous. 2024. AI-Powered Parking Management Systems: A Review of Applications and Challenges. (2024). Retrieved March 19, 2026 from https://www.researchgate.net/profile/Islam-Ibrahim-9/publication/387057857_AI-Powered_Parking_Management_Systems_A_Review_of_Applications_and_Challenge/links/675dc7c9eea8d248be6c2b5e/AI-Powered-Parking-Management-Systems-A-Review-of-Applications-and-Challenge.pdf
- [19] Waheb A. Jabbar, Chong Wen Wei, Nur Atiqah Ainaa M. Azmi, and Nur Aiman Haironnazli. 2021. An IoT Raspberry Pi-based parking management system for smart

- campus[Formula presented]. *Internet of Things (Netherlands)* 14, (June 2021).
<https://doi.org/10.1016/J.IOT.2021.100387>
- [20] Amol Jadhav. 2025. Unified Multi-Platform Messaging System for Seamless Client Communication. *INTERANTIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT* 09, 02 (February 2025), 1–9.
<https://doi.org/10.55041/IJSREM41278>
- [21] Jenny E. Fawei, Oyindinpre Bioko, and Gabriel Ebiowei Moses. 2024. SmartGate: A Biometric Access Control and Attendance System. Retrieved March 19, 2026 from
https://www.researchgate.net/profile/Ebiowei-Gabriel/publication/397689228_SmartGate_A_Biometric_Access_Control_and_Attendance_System/links/691b6b7dd3a2c801362bde86/SmartGate-A-Biometric-Access-Control-and-Attendance-System.pdf#page=38
- [22] Mary Jo, Salvacion Goetsch, and Jonathan E Lobaton. 2023. International Journal of Scientific Research and Management (IJSRM) ||Volume||11||Issue||12||Pages||428-446||2023|| Website: <https://ijsrm.net> ISSN (e): 2321-3418 Characteristics, Roles and Challenges of Traffic Personnel: Implications toward Efficient T.... (2023).
<https://doi.org/10.18535/ijsrm/v11i12.11a02>
- [23] Jing Leng and Yu Guo. 2024. AI-Powered Smart Parking Management: Optimizing Allocation and Safety. *ACM International Conference Proceeding Series* 24, (December 2024), 397–403. <https://doi.org/10.1145/3697355.3697421>
- [24] Marvin Carl May, Daniel Glatter, Dominik Arnold, Daniel Pfeffer, and Gisela Lanza. 2024. IIoT System Canvas — From architecture patterns towards an IIoT development framework. *J. Manuf. Syst.* 72, (February 2024), 437–459.
<https://doi.org/10.1016/j.jmsy.2023.12.001>
- [25] Monika Shah, Deepika Shukla, and Dhara Pandya. 2020. Smart Gate: Intelligent Security System Based on Face Recognition. Retrieved March 19, 2026 from

- https://www.researchgate.net/profile/Monika-Shah-4/publication/348806525_Smart_Gate_Intelligent_Security_System_Based_on_Face_Recognition/links/60489c8a92851c077f2b9287/Smart-Gate-Intelligent-Security-System-Based-on-Face-Recognition.pdf
- [26] Auwal Alhassan Musa, Salim Idris Malami, Fayez Alanazi, Wassef Ounaies, Mohammed Alshammari, and Sadi Ibrahim Haruna. 2023. Sustainable Traffic Management for Smart Cities Using Internet-of-Things-Oriented Intelligent Transportation Systems (ITS): Challenges and Recommendations. *Sustainability* 2023, Vol. 15, Page 9859 15, 13 (June 2023), 9859. <https://doi.org/10.3390/SU15139859>
- [27] Mudrikah Nasyiah, Bayu Kelana, and Anggar Riskinato. 2024. System Usability Scale for Measuring Usability of Social Network Applications from User Perspectives. *E3S Web of Conferences* 483, (January 2024). <https://doi.org/10.1051/E3SCONF/202448303010>
- [28] Gunj Patel, Sunit Parmar, Kavan Patel, Ansh Patel, Darshni Mistry, and Dhara Patel. 2025. RFID-Based Vehicle Security & Surveillance System with Automated Access Control and CCTV Integration. (May 2025). <https://doi.org/10.21203/rs.3.rs-6456771/v1>
- [29] Dwi Priyadharshini, Resman Yamuna, Gunadnya Banu, and Sha Dhas Bensam. 2024. IoT-based Railway Level Crossing Gate Control System. *2nd International Conference on Intelligent Cyber Physical Systems and Internet of Things, ICoICI 2024 - Proceedings* (2024), 293–297. <https://doi.org/10.1109/ICoICI62503.2024.10696415>
- [30] Puji Ratwiyanti, Ahlan Ismono, and Almas Dhiya Rahmani. 2024. Implementation of Equivalence Partitioning Techniques and ISO/IEC 25010 Functional Suitability Standards for Testing SCM Applications in Small-Scale Automotive Industries. (2024).
- [31] Chadatarn Raksawat and Pattama Charoenporn. 2021. Software Testing System Development Based on ISO 29119. (2021). <https://doi.org/10.12720/jait.12.2.128-134>
- [32] Aarthi S Sree and Ug scholars. 2025. IOT-BASED DAM LEVEL MONITORING AND AUTOMATED GATE CONTROL SYSTEM. (2025).

-
- [33] Vijendra Tarune and Devendrac Gautam. 2025. Designing an mLearning Solution for Discrete Mathematics on the Mobile Platform. *Designing an mLearning Solution for Discrete Mathematics on the Mobile Platform* 12, (2025). Retrieved April 16, 2026 from www.jetir.org
- [34] Zaheer Mustafa and Greyson Alwin. 2025. AI-Powered Parking Systems: Enhancing Urban Mobility Through IoT and Automation. (2025). Retrieved March 19, 2026 from https://www.researchgate.net/profile/Greyson-Alwin/publication/389027144_AI-Powered_Parking_Systems_Enhancing_Urban_Mobility_Through_IoT_and_Automation/links/67b0c8bf96e7fb48b9c44aa9/AI-Powered-Parking-Systems-Enhancing-Urban-Mobility-Through-IoT-and-Automation.pdf

CHAPTER 3

TECHNICAL BACKGROUND

This section highlights the technology behind the Integrated Smart Campus Vehicle Management System. This includes the research design, the approach to developing the software component, and the specific hardware and software requirements for the implementation of the system in CSPC.

3.1 Research Design and Methods

This research will involve a Descriptive-Applied Research design with a focus on using a mix of qualitative and quantitative techniques in bridging the gap between existing operational problems and technical solutions [2]. The use of this type of research design is two-pronged; descriptive techniques will enable the documentation of existing manual vehicle monitoring processes and challenges such as difficulties in processing data from the logbook as well as the time required to manually verify sticker IDs, while applied techniques will address implementation issues of the RFID and AI solution to existing operational problems.

For the achievement of the first objective, which is to conduct an assessment of existing manual processes and product backlog, various qualitative techniques can be used. Some of the best approaches include conducting interviews with key individuals from the GSU and campus security as well as conducting observations in real-life traffic flow at CSPC gates. These techniques can be highly beneficial for identifying existing operational bottlenecks in the current manual process.

The qualitative data collected will be subjected to thematic analysis to identify recurrent themes and develop the technical specifications required for achieving the second objective. In order to overcome the obstacles presented by the intricacy involved, as well as the development of the components that make up the core functionalities (User Management, Vehicle Registration, RFID Gate Entry, AI-Parking System, and Violations Logging), an Agile software development 2 methodology will be employed. Instead of using a linear methodology, the Agile process will allow

the team of researchers to break down the project into smaller but more manageable components called “Sprints”.

The third goal will involve an assessment of the system through black box testing (BBT). Here, a quantitative approach will be taken by way of performance tests and structured survey questionnaires. User satisfaction surveys regarding aspects such as functionality, efficiency, and reliability will be conducted using six major factors: Functionality, Usability, Reliability, Security and Performance. At the same time, tests carried out using system performance tests will evaluate how quickly the system responds as well as how effectively it can handle transactions in different levels of traffic in the college campus. Quantitative evaluation will offer proof that will justify the effectiveness and acceptability of the system being assessed.

3.1.1 Data Sources

This research will use both primary and secondary sources of information in order to make sure that the creation of the Integrated Smart Campus Vehicle Management System is in line with the needs of Camarines Sur Polytechnic Colleges (CSPC). The sources for primary data collection will be the direct inputs from the major stakeholders/users of the system, whereas the sources for secondary data will comprise documentation available within the organization as well as technical references.

Sampling Technique

The researchers’ group will employ purposive sampling, which falls under the category of non-probability sampling techniques. In this approach, samples are taken based on certain characteristics of the participants. This method of selecting participants will be used specifically because data should reflect the particular technical requirements of campus security. In contrast to convenience sampling, which is used for gathering general data, purposive sampling is more focused on research [1]. By focusing on those stakeholders who are in charge of verifying stickers and creating logbooks, we would be able to prepare a list of products and system requirements.

Respondents

In this chapter, the participants who will take part in the study will be discussed, along with the participants whose views would add to the analysis of the proposed system. This can be done through involving the selected participants in the following aspects of campus security, vehicle registration, and campus transportation.

Table 1: Respondents

RESPONDENTS	PERCENTAGE	SAMPLE	PERCENTAGE
Campus Security Personnel		10	20%
Administrative Staff (GSU)		5	10%
Vehicle Owners		30	60%
IT Experts		5	10%
TOTAL		50	100%

From the information presented in Table 1, it can be noted that the population for this study is made up of campus security officers, general services unit (GSU) administrators, owners of vehicles, and IT experts, all of whom will collaborate to evaluate the suggested system. The percent distribution of participants was calculated through the formula:

where:

$$P = \left(\frac{f}{N} \right) \times 100$$

P = percentage of the group

f = frequency or number of respondents per group

N = total number of respondents (50)

In the current research, the Campus Security Personnel will be used as the main source of information on the issues encountered during the manual identification of vehicles, gate surveillance, and parking. The expertise held by the Campus Security Personnel will enable the collection of useful information on the weaknesses of the system.

Administrative staff of the General Services Unit (GSU) will play a crucial role in supplying relevant data concerning the process of vehicle registration and database management. These individuals will ensure that the system is aligned with the internal process of managing the data on vehicles.

Vehicle owners, who consist of students, faculties, and staff members, will act as primary users of the system. They will give feedback on usability problems, especially those associated with the existing problems such as traffic jams, parking difficulties, and inefficiency in the manual monitoring process. Last but not least, IT specialists will review the proposed system from the standpoint of its technical architecture and performance.

Secondary Sources

However, all information may not always be readily available from personal engagements with the users; therefore, alternative approaches shall be applied for the verification of the results obtained from the research. These will involve documentation and technical information that will provide adequate proof about the procedures involved in monitoring and securing cars. For the attainment of the first objective, the researchers will study the Daily Vehicle Log Books, the Campus Sticker Registration Forms, and the Gate Incident Logs to assess the number of vehicles and detect security weaknesses [7].

To achieve the second goal, which concerns the design of core components like user management, access gates using RFID Technology, and AI-enabled parking lots, the researchers will use academic sources regarding IoT-based smart parking systems [8] and YOLOv9 algorithm[6].

With this regard, they can ensure that their system design analysis aligns well with engineering principles. To achieve the third goal concerning setting evaluation criteria for BBT, the researchers will use testing metrics and parameters from existing software test standards since these aspects are necessary for measuring the performance criteria of Functionality, Security, and Performance.

Finally, in order to achieve the fourth goal, which involves deploying the proposed solution, the 5 researchers will investigate hardware specifications and layout maps of the university where the system will be deployed.

3.1.2 Research Instruments

This chapter describes the research tools employed in gathering data needed for the creation and assessment of the Integrated Smart Campus Vehicle Management System. The data collection tools described are those that guarantee the identification of the system requirements as well as its assessment using software quality measures as presented in Chapter 2.

Survey Questionnaire. The survey questionnaire is an instrument for collecting structured and quantitative data from various stakeholders including students, faculty members, staff, and security personnel. The instrument consists of various specific parts whose objective is the identification of bottlenecks in operations of the current manual vehicle management system employed by CSPC. It employs questions and scales of rating to capture the issues faced by stakeholders such as congestion at gates, inefficiency in checking vehicles for entry, and error in manual recording of the data. By employing this instrument, the researchers are able to capture a broad population of respondents to prove the bottlenecks, which forms the foundation upon which system requirements will be set.

Data obtained from this instrument forms the basis for achieving Objective 1, since the information obtained is used to develop system requirements such as the integration of RFID for access control and AI for parking management. This tool also forms the basis of conducting a comparative analysis between the old manual system and the proposed system.

Black Box Testing (BBT) Matrix. The BBT Matrix is a technical logging device that will help keep track of how the system performs based on Objective 3. This particular instrument is built like a verification table with a list of tests used to check how functional, secure, and resilient the system is to any errors. The recording instrument is built to track input-output actions where 6 specific inputs are related to their outputs without looking into the system code at all. This particular

instrument is set up to accommodate the use of Equivalence Partitioning. Specifically, there are two different columns where we would classify what data is valid and what data is not, including unauthorized and authorized RFID devices as well as right and wrong login credentials.

ISO/IEC 25010 Evaluation Instrument. This test is a top-notch tool for quality assessment using the ISO/IEC 25010 standards framework with an organization structure meant to assess the effectiveness of the implemented system. The test is set out in a format of a standard rubric which measures the five important criteria of quality: functional suitability, usability, reliability, performance efficiency, and security. The test is evaluated based on the five point Likert Scale. The test will be distributed to two types of respondents: IT specialists and users who will provide information about the effectiveness of the implemented software from their perspectives. The use of two different perspectives will ensure that a lot of data about the effectiveness of the implementation of the system is obtained.

3.1.3 Data Gathering Procedure

The data collection process involves the adoption of a well-structured framework that is geared towards meeting the objectives of the research and includes four main stages namely; requirement analysis, system development, system testing, and system evaluation. This method helps to systematically collect and analyze both qualitative and quantitative data to help evaluate the design of the new system that will be proposed. First, requirement analysis involves using a questionnaire for the collection of data from a wider number of participants such as students, teachers, and other staff members.

These questions will be centered around several problems including delayed entry of vehicles, complications during logbook filling, and lack of live tracking in the current system. Data collected from the participants will be analyzed using thematic analysis for qualitative data and descriptive analysis for quantitative data.

After the completion of requirement analysis, the system will be designed and developed through Agile Software Development techniques. The development of the system is performed

through iterative cycles which enables improvement of the system in accordance with feedback from tests. Some of the important components included in the system are user management, vehicle registration, gate access with RFID technology, parking surveillance, and violations tracking.

The software undergoes testing after each developmental iteration through use of the BBT matrix. The phase involves testing the system to evaluate its functionality through running test cases where the expected output from the test case is compared to the actual output through the use of the pass [1] or fail [0]. Equivalence partitioning is used as a tool in testing the functionality of the system in terms of valid and invalid inputs. Other tests carried out include system performance and system security testing. For this, tools such as Apache JMeter and OWASP ZAP are applied.

System reliability testing is conducted to evaluate how well the system performs when operating continuously without interruptions. In the next step, the developed system prototype is evaluated by selected respondents which include students, faculty, staff, and IT experts among others. The participants are requested to evaluate the system after testing and complete an evaluation form for the system according to ISO/IEC 25010 guidelines. Five-point Likert scale is used to evaluate the performance of the system based on five aspects that include: functional suitability, usability, reliability, performance efficiency and security.

The last step in the process entails the integration of qualitative information obtained through interviewees and observations with the quantitative information obtained through system testing and system evaluation. This is to make sure that both the technical and evaluative aspects are taken into account while making judgments on the performance of the system. The combination of information sources adds credibility to the study and serves as a strong foundation for the decision of deploying the system on the CSPC campus

3.1.4 Data Analysis Procedure

The following methodological approach will be used by the researchers in order to analyze the data that will be collected during this research. The use of such a methodology will guarantee

the objectivity of the conclusions that will be reached regarding the efficiency and safety of the suggested system within the framework of the technical requirements of the Smart Campus setting.

Likert Scale Analysis. This will be achieved through the use of a five-point Likert scale in which each perception will be measured by assigning numerical values depending on the degree of importance attached to them. The scale is defined as follows: 5 – Strongly Agree, 4 – Agree, 3 – Neutral, 2 – Disagree, and 1 – Strongly Disagree.

The computation of the evaluation score for the whole system will be done through the use of the weighted mean technique. This refers to the method of calculating the average score from an individual's point of view on the criteria being evaluated. This can be computed using the formula:

$$WM = \frac{\sum(f \cdot x)}{N}$$

Where WM refers to the Weighted Mean, f represents the Frequency of the responses, x represents the score for each response on the Likert scale, and N represents the number of responses. Interpretation of the computed weighted mean scores is done using a predetermined scale, to determine if the system is acceptable or not. If the mean scores range from 4.21-5.00, the system is described as an excellent and highly acceptable system. Thus, one can safely make a determination that the system has passed all the set requirements.

If the mean scores range from 3.41-4.20, the system is said to be very satisfactory. According to the findings obtained, system 9 operates well, but there is still potential for improvement. When the scores range from 2.61-3.40, they are considered to be satisfactory. The scores suggest that the system is working well, but should be improved. When the mean scores range from 1.81-2.60, the system is considered to be fair. This means that there is a great need for improvement of the system.

The computed mean scores will be interpreted using the following scale:

Table 2: Scale for System Usability Evaluation

RANGE	LEVEL	DESCRIPTION
4.21 – 5.00	Highly Effective	The system is highly efficient, secure, and ready for deployment.
3.41 – 4.20	Effective	The system functions effectively with minor improvements needed.
2.61 – 3.40	Moderately Effective	The system is operational but requires improvements
1.81 – 2.60	Slightly Effective	The system has limited functionality and needs major revisions.
1.00 – 1.80	Ineffective	The system does not meet required standards.

As reflected in Table 2, the following rating ranges shall be utilized in determining the overall satisfaction and acceptability level of the proposed system among the CSPC community. Mean ratings of 3.41 to 5.00, which can be considered as “Highly Effective” to “Effective,” will therefore mean that the system is suitable for implementation since the system adequately satisfies the needs and functional requirements of its target users. The ratings will mean that the system is suitable for offering efficient vehicle monitoring, accurate access control using RFID technology, reliable violation tracking, and efficient AI-powered parking management in the campus environment.

However, in case the ratings fall short of the predetermined rating scale, it would indicate that there is a need for improving some aspects of the system, which could include the response time of the web application, improvement in the parking detection algorithm through the incorporation of Artificial Intelligence (AI), improvements in the security systems employed, enhancements in the user interface, and improvement in stability in varied operating conditions. Moreover, the implementation of the mentioned rating scale would enable researchers to evaluate users' perception quantitatively.

In this way, the strengths of the proposed system, as well as the aspects of the system that still need further development and improvement, can be systematically compared using the various

criteria being considered, which include security, reliability, usability, functionality, and performance. Another advantage of conducting this assessment is the ability to determine whether the proposed Integrated Smart Campus Vehicle Management System will meet the requirements and expectations of the Camarines Sur Polytechnic Colleges.

Table 3: Functional Evaluation Scale

RANGE	INTERPRETATION	DESCRIPTION
0.00 – 0.33	Unsatisfactory	The system does not meet user requirements. Users encounter difficulties in performing tasks, and major revisions are needed.
0.34 – 0.66	Satisfactory	The system meets basic requirements but has minor issues in usability or performance. Improvements are recommended.
0.67 – 1.00	Excellent	The system meets all user requirements. It is easy to use, functions correctly, and is ready for deployment.

The following criteria scale is used in Table 3 to determine whether the system designed will be viable depending on the performance of the system functions according to the requirements of the user. The criteria scale is made up of three types, giving a clear way of evaluating the performance of the system functionalities. In relation to the criteria, a score ranging between 0.67 and 1.00, which can be described as Fully Accepted, means that the system has met all specified requirements, works effectively with no problem and can be deployed successfully.

On the other hand, scores falling within the range of 0.34 and 0.66, referred to as Passes, mean that although the system meets the minimal requirements put forward by the users, there are limitations and inconsistencies with regards to functionality and usability that need to be addressed through additional developments. Similarly, scores ranging from 0.00 to 0.33, referred to as Failed, means that the system has failed to meet all the basic requirements and is not fully operational, hence it makes it hard for the user to carry out operations and needs to be revised and improved significantly.

Table 4: Security Evaluation Scale

RANGE	INTERPRETATION	DESCRIPTION
0.00 – 0.33	Not Secured	The system does not provide adequate security protection. It is vulnerable to unauthorized access, data breaches, or other security threats, and major revisions are needed.
0.34 – 0.66	Secured	The system provides basic security measures but may have minor vulnerabilities and weaknesses improvement.
0.67 – 1.00	Highly Secured	The system provides strong security protection, effectively safeguards data, and prevents unauthorized access or threats.

As presented in Table 4 above, the evaluation scale for the security dimension of the proposed system is intended to assess the capability of the system to protect sensitive data and information against security-related threats, vulnerabilities, and unauthorized access. Three categories have been used to divide the security aspect of the system, in order to give a clear assessment of how well the system can maintain the confidentiality, integrity, and security of user data. The category with ratings between 0.67 and 1.00, which is known as “Highly Secured,” suggests that the system shows high levels of security measures in terms of securing data from any breach and thus can be implemented. However, the category with ratings between 0.34 and 0.66, termed “Secured,” indicates that the system offers satisfactory security but has some areas that require further improvements.

Last but not least, a rating value between 0.00 and 0.33 that is viewed as “Not Secured” suggests that there is a lack of appropriate security measures within the system, making it susceptible to threats, risks, and any potential breach of data. Hence, the system having such ratings would be deemed as inappropriate for use because of the existing weakness in its security features. Through the application of the assessment scale in evaluating the security performance of the

system, the researchers would be able to judge how secure it truly is and whether improvements could be made before implementation.

Table 5: Performance Efficiency Evaluation Scale

RANGE	INTERPRETATION	DESCRIPTION
0.00 – 0.33	Unsatisfactory	The system performs poorly, responds slowly, and cannot efficiently handle expected workloads. Major improvements are needed. breaches, access, data or other security threats, and major revisions are needed.
0.34 – 0.66	Satisfactory	The system performs adequately under normal conditions but may experience delays or reduced efficiency under heavier workloads. Improvements are recommended.
0.67 – 1.00	Excellent	The system performs efficiently, responds quickly, and can handle expected workloads without significant delays or issues.

Table 5 shows the rating scale to be used in measuring the performance efficiency of the proposed system. The rating scale measures the level to which the system responds in providing efficient and consistent performance in terms of expected loads and operational scenarios. The rating scale has been categorized into three different classes, which will be used in measuring the performance efficiency of the system. The system is rated as fully passed if it receives a rating between 0.67 to 1.00 because such a system performs well and can deal with expected loads effectively and respond quickly to any request.

Alternatively, the system will be regarded as passed if it gets a rating of between 0.34 to 0.66 since, even though it performs well, the expected loads pose some problems to it, thus requiring improvement. On the other hand, the system will be rated as failed if it receives a score ranging from 0.00 to 0.33 because, in this case, the system is slow and inefficient in handling expected loads.

Table 6: Reliability Evaluation Scale

RANGE	INTERPRETATION	DESCRIPTION
0.00 – 0.33	Unsatisfactory	The system frequently encounters errors, crashes, or inconsistent results. Major improvements are needed.
0.34 – 0.66	Satisfactory	The system operates under normal conditions but may occasionally encounter minor errors or inconsistencies that require improvement.
0.67 – 1.00	Excellent	The system operates consistently and reliably with minimal errors or interruptions during operation.

Table 6 will serve as the measure scale in determining the reliability test for the newly developed system. This is because the ability of the system to function effectively and without encountering technical issues within the specified duration of its use will be considered as part of the process in making such an assessment. In general, there are three major scales that can be used to rate the performance of the system according to its stability and reliability. The first one ranges from 0.67 – 1.00 (Fully passed). It means that the system functions effectively and encounters very few disturbances. The next one has a rating of 0.34 – 0.66 (Passed), indicating that there are slight problems that may arise. Such difficulties may need correction and adjustments. Lastly, ratings ranging from 0.00 – 0.33 (Failed) mean that there are lots of troubles experienced by the system.

Weighted Mean. This formula is used to calculate the average rating from the respondents. It acts as an indicator to interpret the findings of the survey in numeric form taking into account the number and the associated values of the answers given. The weighted mean assists the researchers in establishing the performance of the system based on the user's evaluation.

The weighted mean is computed using the formula:

Where:

$$\bar{w} = \frac{\sum w_i x_i}{\sum w_i}$$

w^- = Weighted Mean

Σ = Summation

w_i = Weight or frequency of responses

x_i = Value of each response

3.2 Theoretical or Conceptual Background

This section describes the theories and concepts needed for designing and developing the Integrated Smart Campus Vehicle Management System. This includes all important software engineering concepts, architecture patterns, and frameworks needed for integrating the IoT into the system. By laying down the technical foundation, it becomes possible to shift from the current manual management of vehicles to a smarter automated one that adheres to industrial standards.

3.2.1 Software Development Methodology

In this section, the software development approach that was employed in developing the proposed system will be detailed. In implementing this research project, the Agile Development Methodology approach has been employed.

Agile Development Methodology is very appropriate for this project since it provides an environment that encourages the continued development and improvement of new system components, such as the artificial intelligence parking management module, at the same time that existing components can be tested in subsequent development cycles (sprints). Through this approach, a project is broken down into phases during which working components are developed that form part of a whole working system. It also allows continuous testing and improvement, making the system more flexible, reliable, and easier to adapt to the needs of the institution.

The Agile Lifecycle shown in Figure 1 will be the roadmap used in constructing the intended system via the following phases:



Figure 1. Agile Development Methodology

Plan. Project content is created via detailed project definition where the identification of essential components needed by the CSPC organization is carried out. During this phase, the security officers together with the administration work in close cooperation with the researchers for the development of user stories. User stories provide information about essential needs, such as automatic tracking of entry and exits and parking spots' statuses. Another thing, the research team performs an exhaustive feasibility study in order to see how the Laravel Framework can facilitate physical hardware integration into the system. At this stage, a product backlog is developed and sprint planning is performed.

Design. At this point in time, a thorough architectural design documentation is created that will include the overall structure of the system architecture. In the course of work, we are paying great attention to the creation of a user-friendly security interface that will be useful for security guards as well as system administrators. Importantly, at this stage of the design process, it is necessary to develop the database schema with the help of Eloquent ORM as well as define requirement models that will involve the creation of data models and process models for handling

RFID data streams as well as YOLOv9 Vehicle Detection Engine Object Detection Models. System architecture makes use of the MVC architecture pattern that Laravel provides.

Develop. Development entails working hard to convert all the requirements and design specifications into an operational piece of software through vigorous coding work. The scientists use Laravel framework for the actual backend code implementation due to the provision of security, routing, and database management. Even as the website management software is developed using Laravel, Python comes in handy for developing AI components used to analyze video streams. In addition to developing the back end and front end simultaneously, another important activity that happens during this stage is linking of RFID sensors and ESP32 microcontrollers with the actual software.

Test. The system will then undergo rigorous testing soon after its construction in the development phase. The testing phase involves various levels of testing such as testing of the Laravel modules through unit testing (Controller, Model, Middleware) to ensure that AI-powered parking and RFID gate integration work through integration testing, and finally stress testing to measure the performance of the system when there is heavy traffic at the college gates. System testing is done to test the efficiency of vehicle detection based on the RFID chip.

Deploy. The software will then be transferred from the development stage to deployment in the deployment stage, following a test run of one iteration that turns out to be successful. During deployment, the installation of the RFID and AI cameras will take place physically within the specified locations on campus. The server environment for the application to be used will be prepared by the researchers such that it will be accessible online through the dashboard provided by Laravel. It is at this stage that the actual use of hardware and software takes place in monitoring vehicle parking activities on campus.

Review. Review is the final phase of the cycle process, where the team analyzes the outcome generated from implementing the latest iteration relative to its intended goal. During this phase, the team receives feedback regarding the impact of the application on the workflow of the security guards and GSU staff. The differences observed in performance, such as AI delay and RFID distance, will be considered for upcoming iterations. Feedback gathered at this phase can be used to enhance the product backlog of the plan phase.

3.2.2 Design Patterns Algorithms

In the next section, we shall consider the technological architecture and its relevance in building these features. It would be important to highlight that the main attention will be paid to the architecture of the application and the algorithm used for detecting and monitoring the vehicles.

Model-View-Controller (MVC) Pattern. In this respect, the architecture of the software will incorporate MVC architecture which is at the heart of the Laravel framework. The Model part of the MVC architecture will be used to manage all database operations and validation of data related to vehicles and users, thanks to the Eloquent Object Relational Mapper, on the other hand the view part will be used to produce the front-end of the security officers and administrators, using the Blade template engine, while the controller will bridge hardware logic and user interaction.

YOLOv9 (You Only Look Once) Algorithm. The proposed AI-powered Parking Monitoring Module will utilize the YOLOv9 model to detect vehicles from images captured by the campus IP camera. YOLOv9 is an advanced object detection model, which uses PGI or Programmable Gradient Information for reducing the loss of data in deep learning-based feature extraction processes. The advantage of using YOLOv9 is that it will enable the detection of multiple vehicles simultaneously with extremely high speed and accuracy, thereby making sure that the parking lot occupancy information is provided in real time [3].

Intersection Over Union (IoU) Algorithm. As an aid to help the artificial intelligence figure out whether there is any space that can be used for parking, an IoU algorithm is applied. In

the process of applying this mathematical algorithm, the percentage overlap between the object's bounding box and the assigned location of the parking slot is calculated. By using this predetermined overlap threshold (over 50%), it is possible to determine whether a certain parking slot is "Occupied" or "Vacant".

RFID Anti-Collision Algorithm. For making sure that the process of collecting data using the RFID Gate Access Module is accurate, an anti-collision system that comes naturally with the UHF RFID system will be used. In doing so, it will be possible to avoid signal overlap, particularly in cases where there are multiple cars with RFID tags accessing the gate simultaneously. Consequently, it will be easy to confirm the Electronic Product Code in the Laravel backend without any loss of data [5].

Bcrypt Hashing Algorithm. In order to make sure that all the requirements regarding the security of the system are achieved, Bcrypt algorithm will be employed in the User Management Module. The use of the Bcrypt algorithm ensures that the user details of both administrators and users are safeguarded by employing salted hashing and thus no rainbow tables and brute force attacks take place. This proven security protocol will ensure that all the institutional data remains secure in the smart campus.

Observer Pattern. For this, the Observer Design Pattern shall be used to ensure instant communication between the physical hardware and the software installed on the website. In the Observer Design Pattern, Laravel would be acting as the observer that responds to the events created by the subject, including the RFID reader and AI modules. Whenever a new tag is detected or any violation takes place, the system would process all the data to instantly alter the status of the gate.

Repository Pattern. To improve definition of the system architecture under discussion, we will use Repository Pattern, which will make an interface between the layers of data mapping and domain. The repository pattern is meant to keep data access code together, which will improve readability and testing during the Black Box Testing stage. In addition, abstraction of the data layer

will help the research team guarantee that no significant alterations are made to the business logic in case of database schema changes.

3.3 System Architecture

The following sections will provide an overview of the basic design and layout of the proposed system, as well as the interactions among the various components of the proposed design. This information will be presented to highlight the integration of the various processes, data handling, and interfaces in order for the system to operate effectively and securely.

Figure 2 illustrates the architectural design of the proposed system

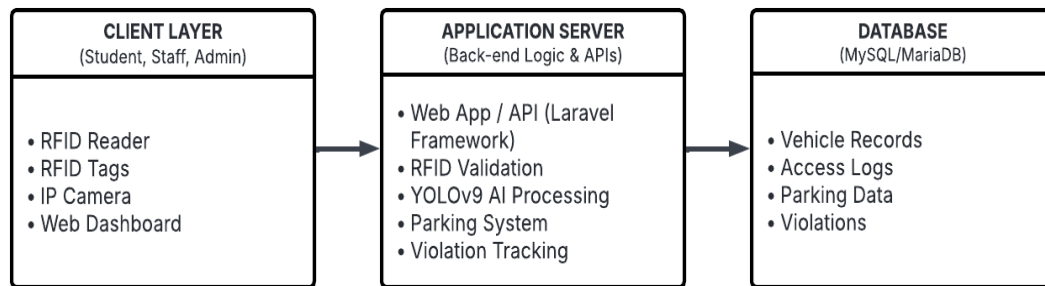


Figure 2. System Architecture

As seen in Figure 2 above, this system will use a three-tier architecture design which separates the user interface, the application layer, and the database layer. With such an architecture, users, who will include CSPC security officials and administrators, will be able to access the system via web clients whereas important operations would take place on the secure server.

Client Layer will be the initial layer for capturing input from the hardware and raw data. The RFID Gate Access Module will capture data with the use of UHF RFID Readers while the AI-Powered Parking Monitoring Module will capture data with the help of IP cameras. Input signals as well as video feeds will be collected at this level and then transmitted to the server. This arrangement will prevent any performance bottleneck arising due to data acquisition and transmission in the real-time system.

The Application Server serves as the main center for the application, which is where all business rules, workflows, and data validations will occur. All of the business logic, including Laravel Backend, YOLOv9 car recognition, RFID credential check, and violations, will be done here. The use of a centralized design in this system ensures that data validation and error checking will always be uniform, thus preventing data corruption when communicating with hardware components.

The Database Server should serve as the data store that would hold all the information related to registrations, access, and logins on MySQL. To fulfill the need of Security, the database should only process and authenticate the requests that have been sent from the Application Server. Database layer would be able to provide assistance to both modules of user management and registration of vehicles during the entire research process, because it has the potential to develop a relational model that develops an audit trail.

Figure 3 shows the pie chart depicting the allocation of the functional components of the system.

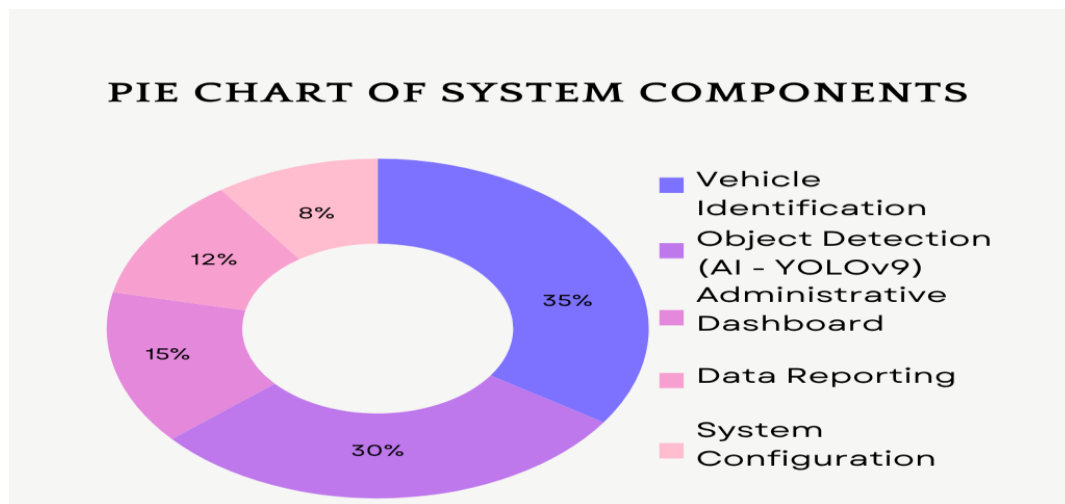


Figure 3. Pie Chart of System Components

It is clear from the above figure (figure 3) that the graph represents the percentage share of various systems in relation to the defined objectives. The highest percentage is taken up by the RFID Gate Access and AI-Powered Parking Monitoring systems. It is worth noting that these systems serve as the technological backbone for this study because they are supported by other systems such as Violation Tracking, User Management, and Registration. Through this approach, the development emphasis has been placed on automation due to the challenges encountered in the early stage of the project.

3.4 Software Development Stack

In this section, the technologies employed in the development of the proposed system are discussed. The parts of the development environment that help achieve system functionalities are emphasized. This section guarantees that the development of the proposed system uses proper software technologies that facilitate the implementation of the system.

Table 7 presents the software development stack specification used in building the proposed system.

Table 7: Software Development Stack Specification

COMPONENT	TECHNOLOGY	VERSION
Operations System	Windows 10/11	22h2/Pro
Web Server	Apache	2.4.58
Back-end Language	PHP (Laravel Framework)	8.2
Database Server	MariaDB	10.4.32
AI Inference Engine	Python (PyTorch)	3.10
Scripting Language	JavaScript (ES+)	Vanilla
Development Tool	VS Code	Latest

As can be observed from Table 7 above there are certain software elements employed within the proposed system to ensure its efficient development and deployment. First, the proposed system is deployed based on the WAMP stack architecture, which is a combination of Windows, Apache, MariaDB, and PHP software elements. It provides a reliable platform for the deployment of sophisticated web applications. In the case of the proposed system, this will ensure compatibility

among the different software elements, namely the operating system, the web server, relational database, and scripting language, to ensure consistency of operations.

Within this technical stack, PHP, running with the Laravel framework, will be responsible for driving the administrative functionality of the application. This performs a number of administrative operations such as checking the credentials of the users to gain access into the gateways, real-time updates to the database about parking space availability, and auditing through Eloquent ORM. The ability to use Laravel's middleware and the use of the built-in authentication features provide an additional level of security in controlling access to certain modules.

With respect to controlling the monitoring functions of intelligence within the project, Python will be used as the AI Inference Engine. With the help of PyTorch and the YOLOv9 algorithm, this feature will process live video streams in order to identify whether any vehicles are present on campus as well as whether their parking spaces are occupied. Information about the vehicles detected using the Python interface is then shared with the Laravel back-end. For the purpose of establishing contact with the various nodes on campus and sending requests, the Apache web server acts as the interface, facilitating fast and reliable access to the administrative dashboard.

The relational database engine used in the system is MariaDB, which is designed to work with the relational data of the registered users, vehicles, and entry times. Vanilla JavaScript is used in the front-end development process to ensure that the updates and interactions with the security dashboard are done in real-time without having to reload the entire page.

3.5 Hardware Development Stack

This chapter will discuss the hardware components needed for the successful implementation of the proposed system. It will present the computing capabilities needed to guarantee efficient system performance. The purpose of this chapter is to ensure that the proposed system works efficiently through the use of existing hardware facilities.

Table 8 presents the hardware development stack specification utilized in the system development process.

Table 8: Hardware Development Stack Specification

COMPONENT	TECHNOLOGY	VERSION
Development Machine	Intel Core i7 / Ryzen 7	11th Gen / 5000 Series
Mamory (RAM)	DDR4 RAM	16 GB
Graphics	NVIDIA GeForce RTX	30-series (4GB VRAM)
Storage	NVMe M.2 SSD	1 TB
Network Device	Gigabit Ethernet Router	1000 Mbps
Controller	ESP32 Microcontroller	Dual-Core 2.4GHz Wi-Fi/BT
Identification Hardware	UHF RFID Reader	860-960MHz (4-port)
Capturing Device	IP Bullet Camera	4MP /H.265

As illustrated in Table 8, the hardware development stack includes hardware components that offer the necessary computational capability and high bandwidth needed by the system. It is necessary to have a development machine with high computational capacity and 16 GB of DDR4 RAM powered by an Intel Core i7 or an equivalent processor for ensuring a reliable local server while performing inference using artificial intelligence technology. It is imperative to have an NVIDIA GPU included in the development stack because it offers the capability for performing object detections by the YOLOv9 model through CUDA cores, and it takes milliseconds to perform such functions.

In addition, special devices like UHF RFID readers and IP bullet cameras have been introduced to aid in the physical sensing of the campus environment. ESP32 microcontroller is employed to serve as an intermediary between the digital logic system and the physical gate, processing the signals received by the RFID readers and the automated gate motor. A gigabit Ethernet router is introduced to offer enough bandwidth for live HD video feed from the cameras into the Python logic system.

This configuration of hardware allows the team to test the real conditions at which CSPC would operate under a developmental environment. This ensures that all the software elements, particularly the Laravel backend and the inference engine, are calibrated properly based on the

exact response time and detection range of the institutional hardware. Through the use of an NVMe SSD, the hardware is able to effectively perform multiple read and write processes from long video logs as well as the database.

3.6 Tools and Platforms

This part of the chapter provides information about the development tools used for design, implementation, and testing of the proposed system. The tools help to write code effectively, manage databases, integrate the systems, and work in collaboration on the development project. This part helps facilitate the development of the proposed system effectively.

Table 9 presents the development package specifications used in the implementation of the proposed system. process.

Table 9: Development Package Specification

PACKAGE / LIBRARY	VERSION	LIBRARY
Laravel Framework	Latest Version	PHP framework used to develop the system architecture
Bootstrap	5.3	CCS framework for responsive front-end processing
OpenCV	4.8	Open-source library for image and video processing
PySerial	3.5	Python library for communication with RFID hardware

From the data provided in Table 9, the proposed system adopts a number of development packages and libraries that will improve the performance and UI design of the system. VS Code acts as the main IDE because of its lightweight nature, extension capabilities for PHP and Python coding languages, and the ability to run Laravel Artisan via a built-in terminal window. The version control shall be carried out using Git. The use of Git allows the researchers to track back any errors that may occur during the changes on the code.

In the process of developing and simulating the local environment, XAMPP will be used in order to provide a local web server using Apache along with MariaDB. This will help us carry

out some tests on the Laravel modules and queries, the same server where we are going to do in the end at CSPC. PhpMyAdmin is used as part of the project to facilitate database management.

Additionally, front end tools like Bootstrap ensures that the admin panel is fully functional irrespective of whatever platform you may be using whether it's the computer screen in the security office or even the tablet during field operations. To ensure that there's no difference between AI surveillance and reporting on the web, the Python modules known as PySerial and OpenCV will work together to capture the feed from the camera and deliver signal alerts to the backend upon identification of any anomaly. All these platforms ensure that every module is tested and developed.

3.7 Testing Plan

This chapter contains details on the testing procedures and methods to be used in evaluating the effectiveness, reliability, security, and efficiency of the recommended Integrated Smart Campus Vehicle Management System (ISCVMS). The testing process has been developed with the Specific Objectives (SOs) of this research project in mind such that all components of the ISCVMS will be properly tested prior to its deployment in Camarines Sur Polytechnic Colleges (CSPC). The following are the types of testing that will be done using the ISO/IEC 25010 software quality standards and Black Box Testing techniques.

3.7.1 Development Testing Plan

Testing and development are conducted to ensure the technical feasibility of the system within the context of the agile software development life cycle. This helps ensure that modules of software, IoT, artificial intelligence, and databases work as expected towards achieving the research objectives:

Unit Testing. This test exercise is geared towards achieving Specific Objective 2 since it concerns the design and development of the Integrated Smart Campus Vehicle Management System. Each individual module of the Laravel Framework, such as User Management, Vehicle Registration, RFID Monitoring, and Violation Tracking, will undergo testing individually to ensure

that all functions perform as expected. The researchers will use the PHPUnit test framework to validate the functionality of specific aspects of the system, such as authentication of users, transaction on the database, form validations, and roles. This approach of testing guarantees that any erroneous input data, database failures, and attempts of unauthorized access are handled by the system without compromising its integrity. This testing will mainly concentrate on the Functionality and Reliability attributes of the system.

Integration Testing. This testing activity aligns with Specific Objective 2 since it tests the interface between the software application and the IoT hardware elements. The research team will test if the RFID reader, AI-powered parking management module, database server, and web application are communicating appropriately and transferring data in real time. The testing will confirm whether the data acquired from RFID tags via the UHF RFID reader is transferred to the database server via the Python program through the Laravel controller for automating gate operations. Additionally, the researchers will test if the YOLOv9 machine learning model correctly analyzes parking capacity and updates the parking information in the database stored in MariaDB.

Stress and Performance Testing. This test is in line with Specific Objective 3 that entails the assessment of how effective and efficient the system is. The researcher will perform stress and performance tests to establish if the system will be able to perform effectively in peak traffic in CSPC. The researchers will use Apache JMeter to simulate multiple simultaneous scans using RFID, multiple users logging into the system concurrently, and multiple IP cameras attached to the Artificial Intelligence Parking Module. Through the process of performing these tests, the response time, throughput, latency, and stability of the system will be established.

Security Testing. Moreover, this test is also aligned with Specific Objective 3 since it determines the capability of the software to ensure user data safety, safeguard the system against any form of unauthorized access, and secure communication within all devices and the server. The researchers shall employ OWASP ZAP to uncover potential flaws such as SQL injection, broken authentication, and any form of unauthorized access within the web application. Furthermore,

network traffic shall be observed using Wireshark to determine the communication that occurs between the RFID devices, cameras, and the server infrastructure. Sensitive data such as vehicle registration and user authentication shall be ensured to be secure during testing.

System Evaluation Testing. The aforementioned testing technique contributes to Specific Objective 3 by conducting an evaluation of the system through the implementation of ISO/IEC 25010 quality assessment guidelines and Black Box Testing methodology. This is done by measuring the quality of the system based on Functionality, Usability, Reliability, Security, and Performance criteria. Evaluation of users' experience will be conducted through conducting surveys for all stakeholders involved in CSPC, including campus security staff, administration, students, and Information Technology experts. These questions will focus on issues such as ease of navigation, layout of the application, its interactive capacity, and overall satisfaction with the system. Data collected through evaluation will be analyzed using the weighted mean computation approach.

3.7.2 System Evaluation

This sub-section provides relevant data on how the effectiveness of the system developed will be determined. In this context, the process of determining the efficiency, functionality, reliability, usability, and performance of the system put into practice will be described against the background of CSPC. It should be noted that the process of evaluation is vital because it will allow determining whether the ISCVMS functions as planned by the specification and achieves all the objectives defined in the research.

Additionally, the process will enable defining whether the system is capable of effectively helping in campus vehicle monitoring, parking control, RFID access control, and violation detection, providing users with convenience at the same time. The findings from this process will further become the basis for potential improvements and compliance with expected implementation standards.

Functionality Testing. Functionality testing should be done to determine if the system has the capability to perform the operations as expected in terms of performance. The test cases used to check the functionality of the system are shown below in Table 10.

Table 10: Functionality Testing

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
RFID Entry System	Swipe an authorized RFID tag at the reader.	Gate servo activates; local monitor displays "Authorized"; SQL logs the entry.
Gmail OTP Delivery	Request a 2FA code during the system login process.	A unique 6-digit code is delivered to the registered Gmail account via SMS.
Manual Gate Override	Trigger the "Gate Open" command via the web dashboard.	ESP32 receives the network packet and triggers the gate immediately.
Status Toggle Logic	Update a specific vehicle's status to "Restricted" in the database.	System denies entry to that tag; dashboard triggers a visual alert.
Unauthorized RFID	Swipe an unregistered or "unknown" RFID tag.	Gate remains closed; local monitor displays "Access Denied"; entry is logged.
Search Functionality	Input a specific plate number into the search bar.	The system filters the display to show only logs associated with that plate.
Duplicate Tag Entry	Attempt to register a vehicle with an existing RFID serial.	The system blocks registration and displays a "Duplicate ID" warning.
Real-time Monitoring	Monitor the dashboard while a vehicle enters.	The log table updates instantly via AJAX without a manual page refresh.
Field Validation	Submit a registration form with empty required fields.	The system prevents submission and highlights missing fields.
Report Generation	Export the daily logs to a CSV or PDF file.	A file is generated containing all correct timestamps and vehicle data.

The test cases that have been set for functionality tests have been illustrated in Table 10 above, and these play an important role in evaluating the effectiveness of the different elements in the proposed system. It is important to note that the use of test cases in functionality tests plays a crucial role in identifying the efficiency of each element of the proposed system. The application of functional testing entails the ability of the system to process data, perform operations, and produce valid output results.

The testing procedure will include an assessment of the major modules in the system, in order to determine the proper functioning of each of the modules. Some of the essential functions that shall be tested for the system include multi-factor authentication, hardware testing based on ESP32 and RFID technology, machine learning parking monitoring, and management of the vehicles. Tests related to the multi-factor authentication will be done to see if the users can access the system according to their designated role. The hardware testing will be carried out in order to check the communication between ESP32 microcontroller, RFID sensor, gates, and the Laravel Web Application to check for the detection of RFID by the system and correct functionality of the gates. Moreover, the tests relating to AI-powered parking monitoring will see if the parking occupancy is detected and added in the system database effectively.

The testing of the functionality of the system is going to be achieved through the implementation of different scenarios that involve inputs that could either be either valid or invalid. If the input happens to be valid then the test will be aimed at determining whether the software behaves as required and generates the right outputs or not, otherwise if the inputs are invalid, then they will assist in finding any anomalies in the system. The testing will focus on the system behavior when presented with wrong data.

Using these approaches, the researchers will be in position to evaluate the effectiveness, reliability, and consistency of each of the system modules. In general, the function test cases provided in Table 10 will have an important part in verifying the operation of the suggested system as well as its ability to provide campus vehicle management and access control at CSPC. These test cases will also help identify possible errors, system inconsistencies, and performance issues that may affect the overall functionality of the system. Furthermore, the results of the functional testing will serve as a basis for improving system performance and ensuring that all features operate according to the specified technical requirements and user needs.

Usability. Usability tests should be carried out in order to establish how easy or convenient it is for the user to use the system. Below is Table 11 illustrating the usability test scenarios that will be undertaken in order to assess the usability of the system

Table 11: Usability Testing

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
User Satisfaction	Users evaluate their overall satisfaction with the system interface after completing tasks.	Users report a positive experience with a mean Likert score of 4.0 or higher.
Ease of Use	Users interact with the system dashboard and perform basic operations.	Users feel the system is easy to use and understand without assistance.
Interface Clarity	Users assess whether menus, buttons, and labels are understandable.	Users feel the interface is clear and logically organized.
Navigation Comfort	Users navigate between different modules such as logs, settings, and reports.	Users feel comfortable navigating the system with minimal confusion.
Visual Appeal	Users evaluate the color scheme, layout, and overall design.	Users perceive the system as visually pleasing and professional.
User Confidence	Users perform tasks such as RFID search and record access.	Users feel confident and secure while using the system.
Responsiveness Perception	Users observe system reaction time during interactions.	Users feel the system responds quickly and smoothly.
Error Feedback Satisfaction	Users trigger invalid inputs (e.g., duplicate RFID entry).	Users feel that error messages are clear and helpful.
Readability	Users view dashboards and data (logs, parking status, alerts).	Users feel that text and information are easy to read and understand.
Overall Experience	Users complete common system tasks (view logs, generate reports).	Users describe the system as efficient, reliable, and user-friendly. .

The test cases required for the conduct of usability testing have been generated as presented in Table 11 and are key to testing the user-focused design and easy usability of the various components of the recommended system. It is essential to state at this point that the adoption of test cases in usability tests is crucial to assessing the potential effectiveness of navigating through the interface of the recommended system. Usability testing entails how well the system can display information to the user, reduce errors on the side of the user, and provide smooth user-software interaction.

The test scenarios would be used to assess the performance of the major interface components including the administrator's dashboard, real-time logging tools, and reporting tools. Learnability will be assessed in order to determine whether the users are able to interact with the system despite minimal training. Meanwhile, the interface design will be tested to see whether the navigation process as well as information search is efficient while the mobility aspect will be assessed as well.

For usability testing, a procedure to assess the degree of satisfaction and user experience will be used by means of a 5-point Likert Scale questionnaire. The feedback received from users will assist in determining how easy it is for users to comprehend, use, and interact with the system. High levels of feedback will denote the ease with which tasks are carried out while low feedback may mean the existence of possible problems concerning the user interface design, navigation, response time, and user interaction. The assessment will be concerned with the users' perception about the system layout, looks, functionalities, and task completion.

Moreover, through usability testing, the researchers will be able to identify any inconsistency in the areas of UI/UX, formatting, navigation, and other issues that may be encountered by users while using the system. Usability testing will also be useful in assessing whether the system interface is user-friendly, visually appealing, responsive, and understandable for campus security officers, university administrators, and car owners. The importance of the usability test cases provided in Table 11 cannot be underestimated when evaluating the usability, acceptance, effectiveness, and efficiency of the system since they will provide the researchers with the information needed to find out the problematic areas that may be encountered while operating the ISCVMS system. Furthermore, these test cases will provide the researchers with valuable insights into whether the requirements set have been met and whether the ISCVMS system provides efficient operation according to the established standards.

Reliability. Reliability testing is done in order to check whether the system will carry out the intended functions in a consistent manner. The test cases for the reliability testing of the suggested system are presented in Table 12.

Table 12: Reliability Testing

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
24-Hour Soak Test	Run the system continuously for 24 hours under active load.	No system lag, reboots, or database connection timeouts occur
Network Auto-Recovery	RecoverySimulate a WiFi signal drop for 120 seconds while active.	ESP32 automatically reconnects and synchronizes with the dashboard.
Data Integrity	Execute 100 rapid RFID swipes in succession.	SQL database records 100% of entries with precise, non-overlapping timestamps.
Power Cycle Resilience	Manually cut and restore power to the gate controller hardware	Hardware reinitializes and returns to "Ready" state within 10 seconds.
Sensor Persistence	Disconnect the DHT11 sensor while the server is operational.	The system continues to run and logs the error without hardware freezing.
Browser Session Stability	Leave the dashboard open on multiple devices for 8 hours.	Browser sessions remain active and data continues to sync without crashing.
SMTP Service Uptime	Trigger 20 OTP requests over a 1-hour interval.	All 20 emails are dispatched and received without service interruption.
Memory Leak Validation	Perform 500 web requests to the ESP32 in a short duration.	Free heap memory returns to original levels after requests stop.
Hardware Heat Resistance	Operate the gate controller in a non-ventilated enclosure for 12 hours.	No thermal throttling or component failure occurs during operation.
Database Concurrency	Simulate simultaneous data writes from two separate ESP32 units.	SQL database manages both entry logs without data corruption or deadlocks.

For the purpose of evaluating the reliability, stability, and overall performance of the hardware and software components involved in the proposed system, the test cases to be employed in reliability testing have been presented in Table 12. It is necessary to conduct reliability testing because it helps ascertain whether the system can successfully and continually perform the intended processes and activities under both normal and stressful conditions without undergoing performance issues or failure. Test cases prove valuable in determining if the modules in the

proposed system can run efficiently and effectively amid various environmental stressors, continuous use, and complex systems processing tasks. Furthermore, through reliability testing, one can evaluate how well the system is able to ensure uptime, effectuate appropriate recovery techniques, and guarantee data consistency/integrity when faced with unforeseen situations like network outage, hardware failure, and power fluctuation.

Test cases used to validate the reliability of the system are designed to test its stability, fault tolerance, and continuous operation capabilities under diverse conditions. The test cases concentrate on core components such as ESP32 connectivity stability, database writes, and SMTP email services connection. Such an approach is intended to prove that the system remains stable and functional, despite any variations in load, connectivity, and operational scenarios. Moreover, tests for the network auto-recovery capability will also be performed to check whether the system is capable of reconnecting to services and continue working without any additional intervention from an operator. This way, reconnection capabilities between IoT components, server, and other third-party services will be verified.

Moreover, the stability of the hardware components is going to be tested using soak test and environmental stress testing. The reason behind these tests is that it will help in understanding the performance of the system when used over a prolonged period and in difficult conditions which are often observed during peak campus times. Additionally, a test of the performance of the system at the time of multiple database writes will also be carried out to understand how efficiently the system manages multiple data entry operations without any corruption or lag. In general, the reliability test cases listed in Table 12 are very important in understanding whether the system is resilient, stable, and ready for sustained use within the campus environment.

Security Testing. Security testing is carried out in order to test the capability of the proposed system to provide adequate security from any attack. The following table shows the cases

for security testing that will enable the determination of whether the proposed system provides adequate security or not.

Table 13: Security Testing

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
Inactivity Timeout	Leave a logged-in dashboard session idle for 1801 seconds.	Session is destroyed; system forces a redirect to the login screen.
Input Sanitization	Enter SQL malicious strings (e.g., ' OR 1=1) in search fields.	Input is sanitized; the system prevents any unauthorized database queries.
Privilege Escalation	Attempt to access restricted setting files via direct URL manipulation.	The system verifies the current session level and denies unauthorized access.
OTP Replay Defense	Attempt to use an MFA code that has already been verified.	The system rejects the code and requires the generation of a new OTP.
Cross-Site Scripting (XSS)	Inject <code><script>alert('XSS')</script></code> into a vehicle registration text field.	The system escapes or strips the tags; the script does not execute in the browser.
Password Hashing	Inspect the database users table directly for plain-text credentials.	Passwords are stored as non-reversible cryptographic hashes (e.g., Bcrypt/Argon2).
OTP Expiration	Input a valid Gmail OTP code that was sent more than 10 minutes ago.	The system identifies the code as expired and denies the login attempt.
Directory Traversal	Attempt to access system root files by appending <code>../../etc/passwd</code> or similar to the URL.	The web server blocks the request and returns a "403 Forbidden" or "404 Not Found" error.
Brute Force Mitigation	Perform 10 consecutive failed login attempts on a single account.	The system triggers a temporary lockout or displays a CAPTCHA/delay mechanism.
Broken Access Control	Use a valid session token to try and delete a vehicle record without the required permissions.	The backend validates the user's role and rejects the unauthorized deletion request.
Security Vulnerability Scanning	Perform automated security scanning of the web application using OWASP standards.	System identifies and mitigates vulnerabilities such as XSS, SQL injection, and broken authentication using OWASP ZAP.
Load and Stress Security Testing	Simulate multiple concurrent login attempts and system requests under heavy traffic.	System maintains stability and prevents crashes under high load using Apache JMeter.

A list of test cases needed for security testing has been developed and presented in Table 13 above. It is worth mentioning that test cases for security are very crucial when it comes to

assessing the reliability of the protective measures in the system. Test cases for security are very important as far as establishing the ability of the system to protect itself against various attacks is concerned. Security testing refers to the ability of the system to deal with malicious data and also protect stored credentials.

The test cases that focus on the security aspect are meant to be used for testing some of the major components of the system's defense architecture like the multi-factor authentication (MFA) through Gmail, session handling, and input validation filtering mechanisms. The cases that deal with the inactivity timeout and privilege escalation are meant to test the capability of the system in ensuring that the user cannot access areas of the system outside his or her permissions. At the same time, tests on input sanitization and cross-site scripting (XSS) prevention are meant to test the accuracy of data filtering performed by the system.

It should be noted that the test cases will include checking the response of the software to malicious and non-standard input from users. The result of such testing will serve as a starting point for determining the system's defensive reaction, when successful blocking of the attempts means that the security operations are performed correctly by the software; otherwise, the flaws in the security operations can be determined. In addition, the implementation of such tests based on the OWASP standards will allow determining such potential weaknesses of the application as a broken access control or sensitive data exposure.

As a conclusion, it should be admitted that the proposed security test cases provided in Table 13 play an important role in evaluating the system's resistance. The test cases make it possible to assess the work of specific modules in the situation of security threat. It allows defining whether the system works efficiently.

Performance. The system will undergo performance testing to ascertain whether the system can retain its performance and stability at different traffic load scenarios. Table 14 below shows

the performance testing scenarios that will be used to assess the effectiveness and speed of processing by the Integrated Smart Campus Vehicle Management System.

Table 14: Performance Testing

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
Concurrent Request	Simulate 50 simultaneous users accessing the vehicle logs.	Physical response time remains consistently below 1500ms.
Execution Latency	Trigger the gate command via the web interface.	The physical hardware begins movement in less than 500ms.
JMeter Throughput	Execute 50 requests per second targeting the database search.	The system processes all requests without 500-series server errors.
Resource Monitoring	Monitor ESP32 Free Heap during peak web traffic.	Memory remains stable above a 10% threshold; no WDT resets occur.
Boot-up Speed	Measure time from hardware power-on to full WiFi and database connection.	The system is fully operational and in a "Ready" state in under 15 seconds.
Search Latency	Query a database containing over 1,000 historical entry records.	Filtered results are retrieved and rendered on the UI in under 1.2 seconds.
Asynchronous Update	Measure the time for a physical RFID swipe to appear on the live dashboard.	The new entry appears on the monitor in less than 800ms.
Report Generation Load	Monitor dashboard responsiveness while generating a large PDF/CSV report.	The UI remains interactive and does not hang during the file generation process.
Bandwidth Optimization	Measure the data packet size of a single RFID status update.	Packet size remains below 2KB to ensure fast transmission over the network.
Peak Load Stress	Gradually increase concurrent JMeter users until the system reaches failure.	The breakpoint is identified; the system fails gracefully without data corruption.

Performance test cases have been generated in Table 14, which are required in determining the efficiency of the fast processing power of the modules of the proposed system. It must be remembered that performance testing cases are very essential in finding out if the proposed system can withstand the peak hour operations in the CSPC campus. Performance testing is determined by the system's ability to handle multiple connections at one time, reduce hardware latency, and increase server throughput.

The testing test cases will be aimed at testing the basic processing functions like ESP32 communication cycle, web server's request handling through Apache JMeter, and query

optimization on the database side. More specifically, latencies and asynchronic updates will be tested for assessing the system's real-time reaction to actions performed by users. On the other hand, throughput and simultaneous requests will be tested to assess the scalability of the web backend, whereas resource utilization and system start-up will be tested from the viewpoint of the internal efficiency and recovery time.

As may be seen, in the test cases, there is also an assessment of system behavior when operating in normal or stressed modes. The result of the tests becomes a basis for determining the system limits within which the operation of the software is fast and efficient. For example, the low response time test reflects that all the tasks are completed promptly, whereas the high load stress test reflects that the system can withstand the maximum loads possible according to its architecture. Additionally, testing allows us to identify bottlenecks in the network and database layers of the system. Therefore, it may be stated that the performance test cases mentioned in Table 14 are necessary for evaluating the general speed of operations in the system. Test cases help assess the efficiency of each module when overloaded. As a consequence, it becomes possible to judge whether the system works properly.

Notes

- [1] Essi Isaac Obilor. 2023. Convenience and Purposive Sampling Techniques: Are they the Same. *International Journal of Innovative Social & Science Education Research*, (Jan.-Mar., 2023), 1-7.
- [2] Noah Sieck, Cameron Calpin, and Mohammad Almalag. 2020. Machine Vision Smart Parking Using Internet of Things (IoTs) in a Smart University, *IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops)*, Austin, TX, USA, 2020, pp. 1-6, DOI: <https://doi.org/10.1109/PerComWorkshops48775.2020.9156121>.
- [3] Yuheng Shi, Naiyan Wang, and Xiaojie Guo. 2023. YOLOV: Making Still Image Object Detectors Great at Video Object Detection. *Proceedings of the AAAI Conference on Artificial Intelligence*, (June 2024), 37(2), 2254-2262. <https://doi.org/10.1609/aaai.v37i2.25320>.
- [4] Zhengjun Hao. 2023. Method for Identifying Motor Vehicle Traffic Violations Based on Improved YOLOv Network. *Cloud Computing for Intelligent Traffic Management and Control*, (September 2023), DOI: <https://doi.org/10.12694/scpe.v24i3.2335>.
- [5] Zhengjun Hao. 2023. Method for Identifying Motor Vehicle Traffic Violations Based on Improved YOLOv Network. *Cloud Computing for Intelligent Traffic Management and Control*, (September 2023), DOI: <https://doi.org/10.12694/scpe.v24i3.2335>.
- [6] Chien-Yao Wang, I-Hau Yeh and Hong-Yuan Mark Liao. 2024. YOLOv9: Learning What You Want to Learn Using Programmable Gradient Information. *Computer Vision - ECCV 2024*. https://doi.org/10.1007/978-3-031-72751-1_1.
- [7] John W. Creswell, and J. David Creswell. (2023). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches (6th ed.). *SAGE Publications*.

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- [8] Yash Agarwal, Punit Ratnani, Umang Shah, and Puru Jain. 2021. IoT-based Smart Parking System. *2021 5th International Conference on Intelligent Computing and Control Systems*. May 2021, 10.1109/ICICCS51141.2021.9432196.